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ORIGINAL ARTICLE

Prescriptive analytics systems revised: a systematic literature review from an information systems perspective

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Abstract

Prescriptive Analytics Systems (PAS) represent the most mature iteration of business analytics, significantly enhancing organizational decision-making. Recently, research has gained traction, with various technological innovations, including machine learning and artificial intelligence, significantly influencing the design of PAS. Although recent studies highlight these developments, the rising trend focuses on broader implications, such as the synergies and delegation between systems and users in organizational decision-making environments. Against this backdrop, we utilized a systematic literature review of 262 articles to build on this evolving perspective. Guided by general systems theory and socio-technical thinking, the concept of an information systems artifact directed this review. Our first objective was to clarify the essential subsystems, identifying 23 constituent components of PAS. Subsequently, we delved into the meta-level design of PAS, emphasizing the synergy and delegation between the human decision-maker and prescriptive analytics in supporting organizational decisions. From this exploration, four distinct system archetypes emerged: advisory, executive, adaptive, and self-governing PAS. Lastly, we engaged with affordance theory, illuminating the action potential of PAS. Our study advances the perspective on PAS, specifically from a broader socio-technical and information systems viewpoint, highlighting six distinct research directions, acting as a launchpad for future research in the domain.

Keywords Prescriptive analytics · Business analytics · Decision support system · Systematic literature review · Decision-making · Artificial intelligence

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1 Introduction

Decision-making is a cognitive process where individuals select from multiple alternatives. Historically, decisions were primarily based on personal experience, direct observation, or shared knowledge (Santos and Rosati 2015). However, with the widespread use of modern information technology, the interconnectedness of society, and the exponentially growing amount of generated data, decision-making has become increasingly complex. As a result, humans began employing mathematical models and algorithms for advanced decision-making to delegate complex decision tasks to computers.

Against this backdrop, business analytics (BA) research to improve organizational decision-making has gained significant traction. The origin of BA is deeply rooted in operations research (OR), commonly linked with decision support systems. Subsequently, the universal adoption of integrated information systems (IS) has enabled organizations to accumulate substantial quantities of data, culminating in the emergence of concepts such as business intelligence (BI) and, more contemporarily, big data analytics (Mikalef et al. 2018). Despite the field's evolving landscape, the core objective of BA has remained steadfast: to delegate analytical tasks to IS to fortify and streamline decision-making processes within organizations. BA shapes well-informed decisions by providing decision-makers with accurate, comprehensive, and timely information, ultimately driving organizational performance and fostering a competitive advantage in the ever-changing business environment (Holsapple et al. 2014; Mikalef et al. 2020).

Prescriptive Analytics Systems (PAS) embody the most advanced iteration of IS utilized within BA and surpass the capabilities of descriptive analytics, which focus on understanding historical data, and predictive analytics, which forecast the likely future. PAS are designed to guide the best action, considering various factors and constraints to achieve desired outcomes. These systems leverage descriptive and predictive analytics results as inputs, harnessing the insights derived from past events and probable future scenarios to inform their recommendations (Lepenioti et al. 2020).

In recent years, research activity has gained traction, with various technological innovations significantly influencing the design of PAS, foremost machine learning and artificial intelligence (AI), such as deep learning, reinforcement learning, and biologically inspired algorithms (Lepenioti et al. 2020). Survey papers keep track of these developments by classifying and conceptualizing PAS aspects from different perspectives. However, their primary focus is often on algorithmic facets. Further, analytics has been predominantly viewed as a passive tool to be used by the human decision-maker (e.g., Frazzetto et al. 2019; Poornima and Pushpalatha 2020; Lepenioti et al. 2020).

Recently, there has been a noticeable shift in BA research, with researchers and practitioners increasingly examining the broader implications of analytics systems. The synergistic relationship between analytics or AI systems and their users in decision-making processes has emerged as a critical area of investigation (e.g., Rzepka and Berger 2018; Niehaus and Wiesche 2021; Hinsen et al. 2022). The

dynamics in the relationship between analytics tools and human decision-makers are changing drastically, with increased delegation between the two and shifting responsibilities, with analytical systems taking agency and ownership of critical steps in the decision-making process. Specifically, PAS demonstrate a significant bidirectional relationship and an expansive decision-making latitude compared to other analytical systems, which are rather passive, reactive, or anticipatory. Such prescriptive agents can act as human partners or substitutes for behavior-based or outcome-based decision-making (Baird and Maruping 2021). We argue that comprehending these factors within the IS community is imperative for illustrating the integration of algorithmic or technical elements into an overarching view, especially given the heightened interest in PAS and their anticipated expansion in research and practice, necessitating the consolidation of the existing knowledge.

To this end, the IS artifact is an established theoretical framework underpinned by general systems theory (GST) to describe, design, or examine systems in a broader organizational context (Chatterjee et al. 2021). Following socio-technical thinking, an IS artifact comprises two closely interrelated and connected subsystems: the social system, with humans as its central component, and the technical system, encompassing elements such as technical infrastructure, hardware, and software. The subsystems are nested in an open system that receives inputs and produces outputs in an environment (i.e., organizational or industry context). They are in synergy with each other and adaptable, meaning they can change over time. Thereby, IS artifacts are intentionally designed to meet a specific objective (Bostrom and Heinen 1977; Niehaus and Wiesche 2021; Chatterjee et al. 2021).

We endeavor to build on this perspective and advance the understanding of PAS with three research objectives nested along the principles of GST (Chatterjee et al. 2021):

(1) First, we aim to consolidate existing literature on PAS, clarifying the essential subsystems, their constituent components, and their interplay and connection to the decision environment. This effort lays the foundation for future research and is crucial in bridging the current knowledge gap. Understanding these aspects is pivotal for successfully deploying PAS, particularly when considering their integration into organizational decision-making processes.
(2) Second, expanding upon the consistent components, we aim to explore the PAS artifacts from a meta-level perspective. We seek to determine whether the literature unveils recurring archetypes or system designs with distinct characteristics, emphasizing the synergy or delegation between the human decision-maker and prescriptive agent and the degree of adaptation (Baird and Maruping 2021). Here, we fall back on human decision-making to better understand how PAS can support organizational decision-making.
(3) Third, as a final objective of our study, we seek to discern the action potential, defining a technology's capabilities to an individual, organization, or industry for a particular purpose. For this, we fall back on affordance theory, which focuses on the action possibilities arising from the relationship between technologies

and their users, often referred to as technology affordances (e.g., Anderson and Robey 2017; Mettler et al. 2017; Effah et al. 2021).

In summary, presenting this synthesized perspective, we posit that it paves the way for subsequent investigations into the core dimensions of PAS within a GST framework, which is especially important for the IS community, effectively setting a research agenda.

To achieve this, we conduct a systematic literature review (SLR), adhering to established methods in IS research (Cooper 1988; Webster and Watson 2002; vom Brocke et al. 2009, 2015). This approach ensures a comprehensive and rigorous examination of pertinent studies, allowing us to derive meaningful insights and identify knowledge gaps in the field of PAS. The paper's structure follows: Sect. 2 provides the research context, encompassing decision-theoretic foundations, BA, GST, and related studies. Section 3 introduces our SLR methodology, detailing the steps to analyze the relevant body of work. Section 4 presents the results of our review, focusing on the key concepts and findings that emerged from the literature. Section 5 discusses the implications of our results, highlighting potential future research streams and avenues for further exploration. Finally, in Sect. 6, we offer concluding remarks, summarizing the contributions of our study and its implications for the field of PAS.

2 Background

To facilitate comprehension of the core facets, in the following sections, we examine human decision-making, BA's role in improving organizational decision-making, and introducing prescriptive analytics in the GST context as an IT artifact. Subsequently, we assess related studies to identify research gaps and underscore the necessity for further investigation.

2.1 Decision-making

Decision-making is fundamentally a biological process rooted in evolution (Santos and Rosati 2015), and decision theory as a research area focuses on examining human choice-making. This field is typically divided into two interrelated aspects (Slovic et al. 1977): normative theory, which prescribes ideal decision-making behavior, and descriptive theory, which describes actual human behavior. In the context of our research questions, the normative theory is of greater relevance, as it assumes decision-makers adhere to rules for consistent and optimal outcomes under given conditions. In practical terms, specifically in IS research, normative theory aims to develop tools that enhance human decision-making (Straub and Welpe 2014).

In this context, human decision-making generally follows a systematic process. While interpretations may vary across domains, the fundamental structure remains consistent (e.g., Simon 1960; Svenson 1992; Schoenfeld 2010; Darioshi and Lahav

2021; Darioshi and Lahav 2021), also in an organizational setting (e.g., Trunk et al. 2020). The process begins with problem identification, followed by alternative generation, evaluation, selection of the most suitable option, decision execution, effectiveness assessment, and iteration for similar problems. This iterative approach constitutes a continuous learning process, yielding increasingly optimized results over time.

Based on the different interpretations, several authors break down the process into overarching phases (e.g., Ren et al. 2006; Leyer et al. 2020). In our work, we adopt a triphase decision-making process consisting of the stages before, during, and after the decision (refer to Fig. 1). Phase 1, evaluation of alternatives, encompasses problem identification, alternative generation, and ranking. Phase 2, decision-making, involves selecting the most appropriate alternative based on the situation, considering the loss and utility of potential consequences, and executing the decision. Phase 3, adaptation and learning, entails assessing the effectiveness of the outcomes in order to modify behavior for subsequent iterations. The triphase process and its components are crucial in designing a PAS and serve as a guiding framework in the synthesis of this study.

2.2 Business analytics

The rapid data volume growth in recent times made BA and Big Data Analytics (BDA) central topics in IS and e-business research (e.g., Pappas et al. 2018; Mikalef et al. 2020; Jensen et al. 2023). Analyzing extensive and diverse data can offer organizations a competitive edge, help achieve strategic and tactical goals, and enhance operational performance by optimizing decision-making processes (Holsapple et al. 2014; Knabke and Olbrich 2018; Oesterreich et al. 2022; Shiau et al. 2023). However, the sheer volume of big data, coupled with uncertainty and noisiness, renders it non-self-explanatory (Lepenioti et al. 2020). Consequently, extracting value from data necessitates sophisticated techniques, processes, and practices. BA, in this context, is a multidimensional and interdisciplinary concept, drawing on technologies from computer science and engineering, quantitative methods from mathematics, statistics and econometrics, and decision-theoretic aspects from psychological and behavior sciences (Mortenson et al. 2015).

BA can be conceptualized using domain, technique, and orientation (Holsapple et al. 2014). The domain (i) pertains to the context in which BA is applied (e.g., a decision problem in manufacturing). The second dimension, technique (ii), denotes the methods employed to perform an analytics task, such as linear programming or specific AI or ML techniques. Lastly, orientation (iii) characterizes the objective or direction of thought, addressing questions like 'what does analytics do?' or 'why is it performed?' and can be regarded as the central dimension. A commonly utilized taxonomy to illustrate the orientation of BA applications is the categorization into maturity levels, which

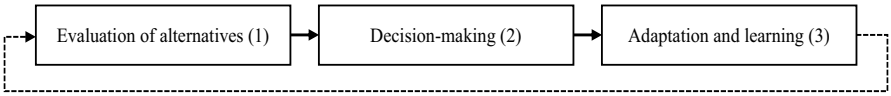


Fig. 1 General phases of decision-making processes

consist of three levels based on their potential and complexity: descriptive analytics, predictive analytics, and prescriptive analytics (e.g., Delen and Ram 2018; Frazzetto et al. 2019; Lepenioti et al. 2020).

The three maturity levels create a synergistic relationship, as depicted in Fig. 2. Descriptive analytics focuses on the past and present by answering questions such as 'What is happening?' or 'What happened?' utilizing traditional BI techniques like Online Analytical Processing (OLAP) or data mining (Delen and Zolbanin 2018). In contrast, predictive analytics anticipates the likely future by addressing the question 'What will happen?' and employs ML methods, such as classification and regression models (Lepenioti et al. 2020). Prescriptive analytics seeks to identify optimal decisions, recommendations, or actions by tackling the question, "What should be done?" (Delen & Ram 2018). This advanced approach employs sophisticated analytics, operations research, and machine learning techniques, including deep learning, mathematical programming, evolutionary computation, and reinforcement learning (Lepenioti et al. 2020).

The full potential of predictive analytics can only be harnessed when combined with prescriptive analytics, which streamlines decision-making processes proactively. Reducing the time interval between event prediction and proactive decision-making is paramount to maximizing business value. Prescriptive analytics generates well-informed decisions based on the outcomes of predictive analytics, considering the most suitable timing for executing actions preceding the anticipated event. On the other hand, descriptive analytics can be utilized after the event to scrutinize its underlying causes and consequences while operating on diverse timescales for reactive or long-term actions. In this regard, the prompt detection of the current state and precise forecasting of emerging events are crucial factors in mitigating potential losses in business value (Krumeich et al. 2016; Lepenioti et al. 2020).

2.3 Prescriptive analytics systems as an IS artifact

The concept of an IS artifact remains ambiguously defined (Chatterjee et al. 2021). However, consensus suggests that it can direct research, clarify

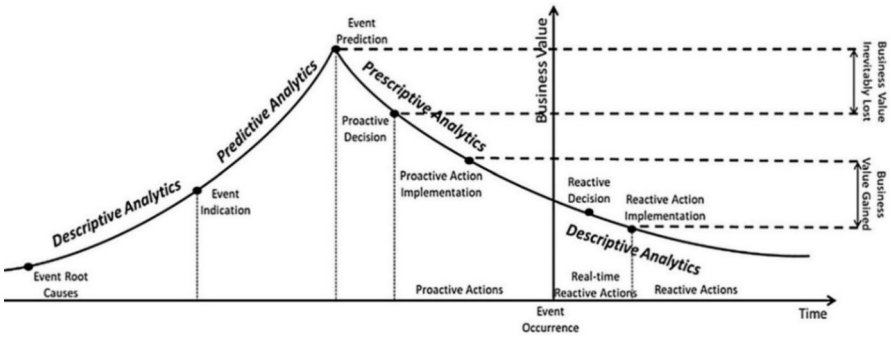


Fig. 2 An overview of the synergies and dynamics of descriptive, predictive, and prescriptive analytics (Krumeich et al. 2016; Lepenioti et al. 2020)

understanding, set boundaries, provide a design framework, and foster novel research perspectives, among other applications (Orlikowski and Iacono 2001; Aier and Fischer 2011; Chatterjee et al. 2021).

One way of conceptualizing an IS artifact is with GST (Kast and Rosenzweig 1972; Chatterjee et al. 2021). Within this theoretic framework, and drawing upon socio-technical thinking, two principal subsystems emerge: the social and the technical (Sarker et al. 2019). The social subsystem is characterized by its components, encompassing individuals with their inherent knowledge, skills, and values, as well as structural facets like organizational hierarchies and reward systems. Conversely, the technical subsystem is described as an assembly of constituent technical components, such as hardware, software, or methodologies, to transmute inputs into outputs, enhancing the performance of an organization (Bostrom and Heinen 1977; Chatterjee et al. 2021). These subsystems interact in synergy, allowing for the exchange of information to fulfill mutual objectives or purposes. Situated as an open system, an IS artifact is embedded within its environment (i.e., in an organizational or industry context), influenced by external factors, and concurrently impacting its surroundings. Central to its design is the adaptability of its subsystems, ensuring stability amid changes (Kast and Rosenzweig 1972; Sarker et al. 2019; Chatterjee et al. 2021).

Further, Chatterjee et al. (2021) underscore the significance of examining the interactions between subsystems using an affording-constraining lens. The affordance theory, introduced initially by Gibson (1986), identifies action possibilities stemming from the relationship between an object and its observer. In IS research, this concept translates to technology affordances, highlighting the potential actions enabled by the relationship between technologies and their users (Anderson and Robey 2017; Mettler et al. 2017; Leidner et al. 2018). Here, affordance can be defined as ”what an individual or organization with a particular purpose can do with a technology” (Majchrzak and Markus 2013), further emphasized by Markus and Silver (2008), who describe them as a user’s interaction potential with a technical object.

The IS artifact as a theoretical concept will guide our SLR, so we elucidate the aspects and their interplay in an exemplary and simplified PAS-supported organizational decision-making problem within manufacturing, precisely, maintenance operations (e.g., Liu et al. 2019; Ansari et al. 2019; Gordon et al. 2020; Wanner et al. 2023), as illustrated in Fig. 3.

At its core, the social subsystem encompasses the individuals involved in the decision-making process, serving as decision-makers responsible for managing and overseeing maintenance processes. Concurrently, the prescriptive agent, as the technical subsystem, comprises the infrastructure, analytics models, and visualization tools to present findings to decision-makers based on inputs from the decision environment. The decision-maker interacts with the technology components. This interplay affords the optimal maintenance schedules to the user, which then can be actioned upon and implemented in the environment, for instance, a production line with multiple machines.

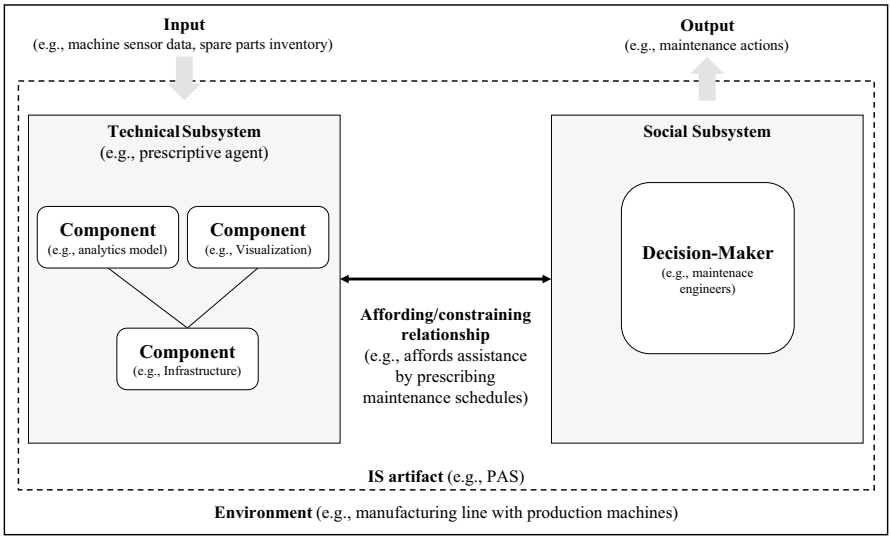


Fig. 3 IS artifact with exemplary PAS-supported decision-making problem (own depiction based on Bostrom and Heinen 1977; Chatterjee et al. 2021)

2.4 Related work

Given the significant research interest in the field, several studies have investigated key topics related to prescriptive analytics. Our analysis emphasizes the importance of our research objectives, and we compile findings, including the purpose addressed in the related work (cf. Table 1). We considered existing systematic and unstructured literature reviews to ensure a well-rounded understanding.

Previous research has primarily concentrated on the technical subsystem of PAS, analyzing its technology components and affordances or applications. For instance, Lepenioti et al. (2020) begin their review with an in-depth analysis of predictive and prescriptive analytics methods before outlining challenges and future directions. Meanwhile, Frazzetto et al. (2019) take a system-oriented approach, emphasizing the various features of PAS, including productivity, infrastructural considerations, and analytical capabilities. Vanani et al. (2021) focus specifically on employing deep learning algorithms in PAS in the Internet of Things, while Stefani and Zschech (2018) provide a conceptualization that considers decision theory as a fundamental aspect. They consolidate various perspectives to derive technology components. Lastly, Poornima and Pushpalatha (2020) adopt an application-oriented approach, providing a comprehensive overview of the usage of PAS in diverse industries. Despite their differences, these studies collectively emphasize the importance of considering various technical factors when developing PAS.

Aside from general reviews of PAS, some studies focus on specific industries or contexts. For instance, Fox et al. (2022) emphasized the importance of maintenance tasks in wind farms and conducted a PAS review specifically for this industry. Soeffker et al. (2022) identified unique requirements for dynamic vehicle

Table 1 Previous literature reviews on prescriptive analytics

Comparison element	Data collection/method (literature reviewed)	Purpose
Lepenioti et al. (2020)	SLR (N = 56)	Synthesis of analytics techniques and methods and proposition of a research agenda
Frazzetto et al. (2019)	Unstructured review	Analysis of technical components for analytics, infrastructure, and productivity
Vanani et al. (2021)	Unstructured review	Prescriptive analytics, specifically using deep learning in an IoT context
Stefani and Zschech (2018)	SLR (N = 30)	Overview of PAS technology components in a coherent view, incorporating aspects from decision theory
Poornima and Pushpalatha (2020)	Unstructured review	Overview of application domains, including analytics methods and infrastructure considerations
Fox et al. (2022)	Unstructured review	PAS for wind-farm operation and maintenance, including perspectives on analytics methods and technology, but also insight into affordances related to the context
Soeffker et al. (2022)	Unstructured review	Conceptualization of decision models, prescriptive methodologies, and computational methods
Bhatt et al. (2023)	SLR (N = 147)	PAS use-cases for sustainable operations research
Kubrak et al. (2022)	SLR (N = 37)	PAS for process monitoring, including aspects such as purpose/objective, data considerations, and analytics techniques summarized in a framework
This Study	SLR (N = 262)	PAS from an IS perspective, constituent components of subsystems, a meta-system level view, and technology affordances

routing and reviewed relevant literature. Bhatt et al. (2023) provided a framework for developing PAS in sustainable operations by identifying five application themes. Meanwhile, Kubrak et al. (2022) explored challenges and suggested areas for future research to enhance the usefulness of prescriptive process monitoring methods. These studies highlight the significance of context-specific considerations and affordances in developing and applying PAS.

In summary, much of the existing research on PAS has been primarily anchored in its technical components and underlying concepts. This review seeks to weave these diverse strands of thought, capitalizing on the foundational works to offer a more holistic perspective. Specifically, we aim to synthesize the current landscape of prescriptive analytics, positioning it as an IS artifact within the broader context of the decision-making process and revealing the delegation of tasks and responsibilities of both the human decision-maker and the prescriptive agent.

3 SLR methodology

In this section, we employ the established SLR methodology in IS research, as outlined by vom Brocke et al. (2009; 2015), incorporating extensions from Cooper (1988) and Webster and Watson (2002). This method consists of five phases: (1) definition of review scope, (2) conceptualization, (3) literature search process, (4) literature analysis and synthesis, and (5) research agenda. Further, we take a descriptive approach to show the current understanding of the literature and reveal patterns, trends, or gaps in current PAS research (Paré et al. 2015).

In this section, we begin by addressing the review scope and conceptualization of the topic, drawing on the theoretical foundations from the previous section. Subsequently, we introduce the literature search process, followed by an initial analysis of the literature sample. The synthesis results will be presented in Sect. 4, while Sect. 5 of this paper will cover the research agenda.

3.1 Scope

To define and present the scope of our SLR, we employed Cooper's (1988) taxonomy with six dimensions. Our (1) focus encompasses research outcomes and applications, including mathematical, conceptual, technological, and infrastructural contributions related to using PAS to understand their aspects better. The (2) goal of our review is to integrate GST perspectives by (3) organizing the results conceptually, following the method outlined by Webster and Watson (2002). We aim for a (4) neutral representation to reveal the current state of PAS-based research. Our review targets a (5) diverse audience, including IS scholars, practitioners, and specialized researchers from the BA community. Lastly, we endeavor to provide a (6) representative coverage of the relevant literature.

3.2 Conceptualization

In our conceptualization, we draw upon the research context and related work and utilize the GST-based IS artifact to guide the organization and structure of our review and its findings. We aim to uncover crucial concepts and elements within PAS to enable a research launchpad. To effectively address our research goals, we will divide our SLR into three distinct foci:

- **Constituent components:** Understanding the constituent components of a PAS is essential. Here, we adopt GST, precisely the notion that IT artifacts can be viewed as transformational models, receiving inputs, transforming them, and generating outputs (Kast and Rosenzweig 1972; Chatterjee et al. 2021).
- **System archetypes:** We aim to explore the meta-system level of PAS artifacts by building on the constituent elements. We are keen to ascertain if the literature reveals recurrent PAS archetypes or designs marked by unique features, spotlighting the delegation and responsibilities between the human decision-maker and prescriptive agent and the degree of adaptation. To achieve this, we lean into the decision-making process as a lens to better understand how PAS are nested here.
- **Technology affordances:** From an IS perspective, we aim to understand the purposes for which PAS are implemented and applied. Focusing our analysis on industry or industry-agnostic use cases and breaking the investigation into specific technology affordances, we will delve deeper into how PAS contribute to and influence decision-making tasks across diverse industry and organizational settings.

3.3 Literature search process

The third phase of the SLR methodology, the literature search process, consists of three subphases: database search, keyword search, and backward and forward search. We outline our procedure as follows (cf. Figure 4).

Initially, we selected interdisciplinary databases such as Web of Science and Scopus, technology-related databases like ACM Digital Library and IEEE Xplore, and AISeL for IS-related outlets. Our search string combined the term 'prescriptive' with several core concepts derived from prior survey articles (Stefani and Zschech 2018; Frazzetto et al. 2019; Lepenioti et al. 2020) (cf. Appendix A for details).

This search yielded 2,597 results (date of search: March 30, 2023). Two researchers collaborated to analyze and screen the papers, using existing conceptual and review papers to establish a shared understanding. Inclusion criteria were set (cf. Table 2) to consider only papers that explicitly address the overall design of PAS or describe specific PAS elements, components, or properties. For example, this includes a diverse spectrum of studies, such as conceptual (Levasseur 2015; e.g., Appelbaum et al. 2017), review (e.g., Poornima and Pushpalatha 2020; Lepenioti et al. 2020), technological/architectural (e.g., Vater et al. 2019; Basdere et al. 2019),

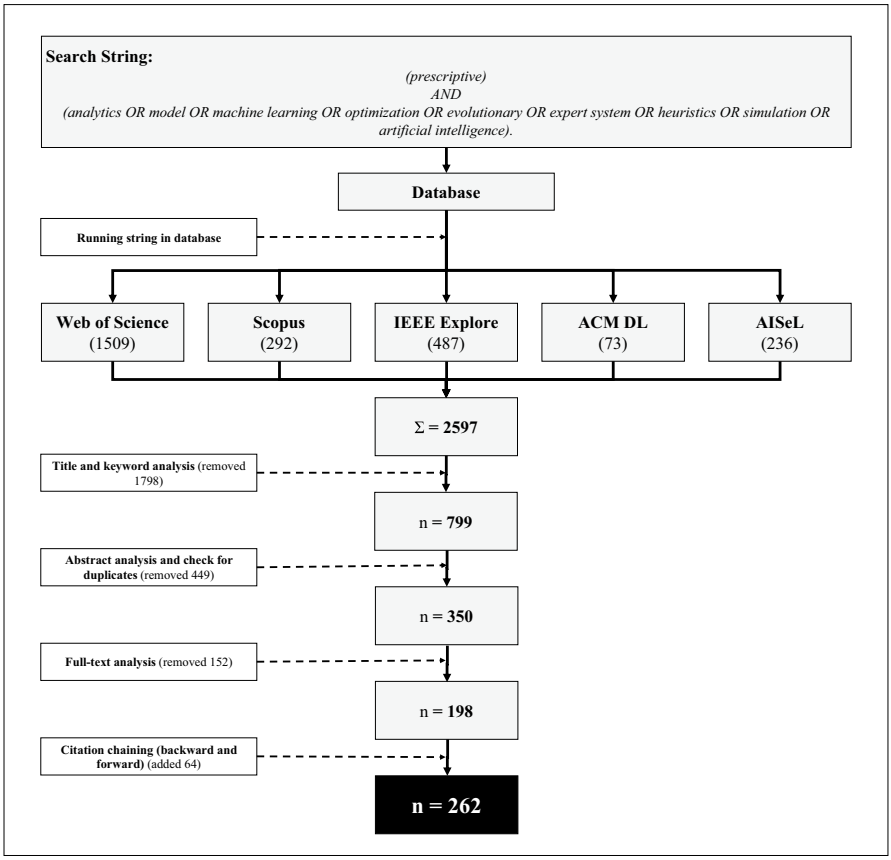


Fig. 4 The literature search process

Table 2 Inclusion and exclusion criteria used in the SLR

Inclusion criteria	Exclusion criteria
• Explicitly addresses the overall design or specific elements, components, or properties of prescriptive analytics systems	• Not written in English • Published before 2010 (and, thus, does not discuss contemporary PAS) • Is a presentation or a research-in-progress paper without significant interim results • Prescriptive analytics is only mentioned without providing more detailed descriptions • Non-academic papers (e.g., practical case studies or reports)

or mathematical papers (e.g., Bertsimas and Kallus 2020; Elmachtoub and Grigas 2022). By contrast, we aimed to exclude papers that only briefly mentioned prescriptive analytics without providing more detailed descriptions (e.g., Swaminathan

2018; Pereira et al. 2021). We also excluded non-academic articles. However, we did not limit our search to only high-ranking journals and conferences to ensure a comprehensive SLR. Webster and Watson (2002) argue that a topic-centric view of the literature is more valuable than a view limited to a few top journals. Further, we excluded articles that were not written in English.

To incrementally exclude irrelevant papers, we applied a stepwise procedure. First, a title and keyword analysis identified 799 relevant articles. After reviewing abstracts and removing duplicates, we removed 449, reducing the number of papers to 350. Full-text screening further reduced the number of relevant articles to 198. To supplement our findings, we conducted citation chaining, both forward search (via Google Scholar) and backward search (via bibliography), adding 64 relevant papers and increasing the total to 262 articles. The complete list of the literature sample is available in Appendix B.

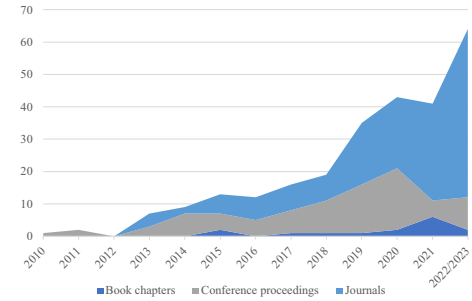
3.4 Overview of the literature sample

Our findings indicate a significant growth in research interest in prescriptive analytics. More than half of our sample was published after 2020, demonstrating the increasing relevance of this area of research. Concerning publication types, approximately 60% of articles are in journals, and 34% of our sample comprises conference papers. A smaller proportion, 6%, is book sections or chapters.

An initial sample analysis allowed us to explore thematic trends by observing the top research outlets. Much of the sample is published in IEEE and ACM proceedings focusing on computer science and technology. Similarly, we observed a clear indication of the operations research and management domain, with publications in journals such as Management Science, European Journal of Operations Research, and others at the intersection between computing, operations research, and industrial engineering (Fig. 5).

In 2021 and 2022, we noticed a considerable drop in conference papers, possibly due to widespread lockdowns in light of the COVID-19 pandemic. Nonetheless, the publication of numerous journal papers during this period contributed to the continued growth of interest in those years. In summary, after initial observation of our

Overview of literature sample, including publication type



Top 13 research outlets

IEEE proceedings (various)	21
European Journal of Operations Research	9
INFORMS Journal on Applied Analytics	9
ACM proceedings (various)	9
Management Science	6
IEEE Access	6
Annals of Operations Research	4
Sustainability	3
Production and Operations Management	3
Expert Systems with Applications	3
Decision Support Systems	3
Computers & Operations Research	3
Computers & Industrial Engineering	3

Fig. 5 Overview of literature sample and top research outlets

literature sample, it is evident that the research is heavily weighted toward technological and mathematical disciplines, which is to be expected given the core of prescriptive analytics.

4 Results

Below, we showcase the findings of the synthesis. We will begin by detailing the constituent components, system archetypes, and finally, the technology affordances of PAS.

4.1 Constituent components

We observed a predominant emphasis on technical components within our literature sample, characterized by its technical focus and the inherent nature of prescriptive analytics. The decision-maker naturally emerges as the pivotal entity in the social subsystem. However, a detailed exploration of the decision processes or structures surrounding PAS-based decision-making is absent in current research, with just a few authors focusing on the decision-maker. For example, Käki et al. (2019) investigate the deviations of decision-makers from model-based recommendations and their impact on the effectiveness of decision-support processes. By examining these discrepancies and discerning the underlying motivations, the authors emphasize the potential for improving the planning process, optimizing model-driven decision-making, and refining the lifecycle management of PAS. Similarly, Caro & de Tejada Cuenca (2023) study the adherence to prescriptive analytics recommendations, highlighting trust as a deterrent, with interpretability as a crucial intervention.

Consequently, the following sections will mirror this emphasis on the technical subsystem and its components. Here, we present a multi-layered concept matrix with 23 concepts. The basic structure of the concept matrix follows GST, that IT artifacts can be viewed as transformational models, receiving inputs, transforming and processing them, and generating outputs (Kast and Rosenzweig 1972; Chatterjee et al. 2021), which we coin "decision formulation", "decision input", "decision processing", and " decision output" in our study. We added "ancillary components" to address additional aspects (Frazzetto et al. 2019) that are not situated in the core of the prescriptive agent but support its integration into organizational decision-making processes and structures.

Table 3 presents a concise summary of the outcomes derived from the concept matrix, featuring exemplary studies corresponding to each concept and the number of hits where the concepts are discussed or mentioned in the text corpus. A detailed overview of all concepts correlated with the literature sample can be found in Appendix C. Further, we added a trend analysis in Appendix H, visualizing the concepts' development across the years.

Each concept (emphasized in bold) is expounded upon in greater detail in the subsequent sections. Further, Fig. 6 integrates and summarizes the findings, offering

Table 3 Concept matrix of constituent components

Constituent components of PAS		Example studies	Frequency
Decision formulation	Decision variables	[7], [19], [28], [48], [53], [69], [110], [117], [176], [243]	147
	Objectives	[4], [27], [45], [56], [76], [110], [146], [166], [195], [262]	158
	Constraints	[8], [33], [45], [72], [92], [117], [152], [185], [217], [243]	137
Decision input	Current state	[15], [30], [53], [67], [90], [105], [130], [163], [200], [236]	133
	Probabilities	[13], [27], [58], [67], [92], [105], [139], [162], [202], [234]	201
Decision processing	Mathematical programming	[10], [35], [42], [74], [97], [131], [152], [178], [204], [233]	149
	Machine learning	[18], [34], [45], [58], [67], [84], [110], [148], [198], [226]	171
	Evolutionary computation	[59], [92], [176], [190], [18], [145], [128], [83], [67], [229]	27
	Simulation	[19], [61], [72], [87], [92], [101], [117], [139], [197], [213]	67
	Logic-based models	[90], [113], [118], [165], [171], [190], [204], [211], [226]	12
	Probabilistic models	[21], [27], [50], [73], [90], [110], [129], [144], [168], [229]	38
Decision output	Single decision	[14], [118], [183], [29], [65], [56], [131], [40], [178], [142]	128
	Multiple decisions	[4], [84], [92], [115], [136], [190], [211], [213], [222], [231]	21
Action mechanisms	Execution	[21], [54], [60], [91], [98], [112], [163], [192], [202], [223]	25
	Adaptation	[16], [40], [60], [79], [92], [123], [162], [194], [196], [220]	34
Ancillary features	Integration	[9], [31], [35], [36], [38], [49], [67], [125], [163], [227]	48
	Distributed computing	[12], [23], [30], [31], [54], [114], [189], [194], [208], [223]	16
	Modulization	[14], [25], [34], [54], [79], [92], [112], [123], [194], [233]	80
	Security- and privacy-preserving	[34], [38], [59], [75], [82], [98], [159], [182], [190], [210]	10
	Workflow interface	[6], [12], [30], [38], [65], [82], [92], [100], [173], [218]	43
	Explainability	[84], [147], [203]	3
	Visualization	[3], [6], [38], [65], [84], [91], [104], [132], [189], [257]	71
	Extensibility	[30], [40], [49], [73], [189], [208], [218], [236]	8

a cohesive representation of the PAS while elucidating the interactions among its core components.

4.1.1 Decision formulation

Decision formulation refers to the essential elements for structuring a decision problem, subdivided into decision variables, objectives, and constraints. **Decision variables** define the object of interest within a decision (Stefani and Zschech 2018). For instance, production planning may involve mapping the manufacturing workforce, machinery, and material flow allocation to each other most profitably (Elmachtoub and Grigas 2022). In this context, the complete set of all potential mappings constitutes the set of all alternatives or competing decisions. When considering in conjunction with the encompassing environmental conditions and various contextual factors, the specification of decision variables plays a pivotal role in delineating specific states and their corresponding outcomes. These states frequently exhibit associations with utility values, encompassing metrics such as costs, profits, or revenues, thus serving as quantitative indicators for overarching **objectives** that demand either minimization or maximization. These objectives are commonly denoted as objective functions, optimization functions, or simply objectives. Additionally, it is pertinent to acknowledge the presence of **constraints**, which often encircle decision spaces,

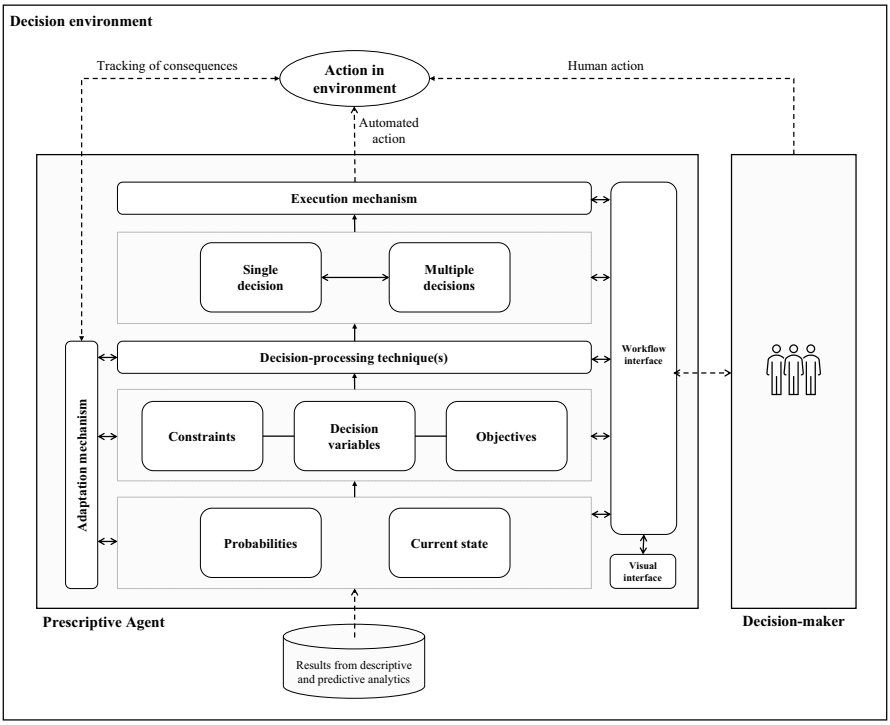


Fig. 6 Exemplary visualization of constituent components of PAS in a coherent view

emanating from natural limitations (e.g., capacity limitations of a machine) or managerial imperatives (Stefani and Zschech 2018; Elmachtoub and Grigas 2022).

4.1.2 Decision input

The input is the foundational component in data-driven decision-making, encompassing essential attributes of analytics processes. While not intrinsic to the core components, numerous authors emphasize these attributes in the context of PAS (cf. Appendix G for an overview). These properties encompass the structural characteristics of the data (Lash and Zhao 2016), its origin, whether external or internal (Bertsimas and Kallus 2020), and the manner of data generation, whether reliant on human-based assumptions, empirical methods or synthetic means (Stefani and Zschech 2018; Ceselli et al. 2019). Additionally, consideration is given to the velocity of data, differentiating between historical and real-time data (Krumeich et al. 2016; Miikkulainen et al. 2021). Subsequently, data undergoes various preprocessing or data engineering steps to prepare the data input for analytical processing (McFowland III et al. 2021), often handed over to descriptive and predictive functions, resulting in the current state and probabilities. Many authors see these preceding analytical results as a foundation of PAS (e.g., Wang et al. 2018; Miikkulainen et al. 2021), thus making them constituents.

Descriptive analytics provide insights into the **current state**, such as patterns or key performance indicators, to assess the existing conditions within the decision context. This information helps identify areas that require modification compared to the current state and serves as a baseline for evaluating the ramifications, e.g., in terms of gains or losses, of a decision (Li et al. 2021). Moreover, it is imperative to recognize that decision problems inherently encompass a degree of uncertainty. This uncertainty may be effectively quantified by utilizing **probabilities** derived from predictive analytics, facilitating an elucidation of the likelihood associated with the impending occurrence of a particular outcome. Probabilities frequently serve as integral components, directly integrated as weightings within the delineation of the objectives within the overarching decision framework (Stefani and Zschech 2018; Wang et al. 2018; Lepenioti et al. 2020; Miikkulainen et al. 2021).

4.1.3 Decision processing

Both the input and formulation frame the decision processing, correspondingly generating prescriptions. Here, the literature refers to various techniques, which can be grouped into mathematical programming, evolutionary computations, machine learning, probabilistic, and logic-based models, which are not mutually exclusive and can be utilized interactively or sequentially, confirming the results of prior literature reviews (Lepenioti et al. 2020). Additionally, we refer to Appendix F for an overview of more detailed techniques and potential subcategories.

Mathematical programming is widely adopted for optimizing objective functions within constrained solution spaces (e.g., McFowland III et al. 2021). In contrast, **evolutionary computations** offer bio-inspired optimization techniques (e.g., Miikkulainen et al. 2021), while **machine learning** (ML) algorithms enable learning

without explicit instructions (Janiesch et al. 2021). Supervised ML facilitates anticipatory decision-making by predicting unseen data (Lash and Zhao 2016). As a subset of ML, reinforcement learning (RL) aims to maximize cumulative rewards in given environments, proving effective for well-formalized decision-making problems (Lepenioti et al. 2021). **Probabilistic models**, such as Bayesian inference or Markov models, calculate event likelihoods and represent causal relationships (Lepenioti et al. 2020). **Logic-based models** examine chains of cause-and-effect relationships leading to specific outcomes (Lepenioti et al. 2020). Lastly, **simulations** enable exploring hypothetical or real-life processes to improve decision-making by generating scenarios and uncovering optimal behaviors for specific situations (Lepenioti et al. 2020).

4.1.4 Decision output

The decision output refers to the result or outcome produced in decision processing. It represents the prescribed course of action or solution from competing decisions based on the decision variable (e.g., all alternative configurations of human resources, machinery, and equipment). While the optimal or **single decision** is typically required, authors emphasize the importance of making the alternatives or **multiple decisions** transparent to the decision-maker and accommodating more complex situations, such as dynamic environments. For example, the prescriptions are tailored to sequential process stages, varying time points, or ever-changing states, ensuring a more adaptive response (e.g., Liu et al. 2019; Brandt et al. 2021).

4.1.5 Action mechanisms

Following the decision-making process, the human decision-maker traditionally performs the subsequent actions within the decision environment. In this context, it is essential to emphasize that the implications and outcomes are detached from the technical subsystem, which primarily functions as a passive tool at the user's disposal. However, recent advancements in AI and ML have triggered a notable shift in IS research (Baird and Maruping 2021). This shift acknowledges the agency of the IS artifact and promotes synergistic collaboration between the technical and social subsystems. In the context of PAS, this transition necessitates the development of specific mechanisms to facilitate action execution, tracking, and adaptation, enabling the technical components to engage in the decision environment autonomously.

Implementing **execution** mechanisms is essential to facilitate actions within the decision environment, distinguishing between two primary execution modes: (i) autonomous execution, where the prescriptive agent independently carries out the decision (e.g., Mazon-Olivo et al. 2018; Soroush et al. 2020), and (ii) execution with human intervention, which may require human confirmation of the decision output before implementation (e.g., Rizzo et al. 2020). As automation levels increase, the importance of tracking becomes more pronounced to document the actions taken in the decision environment, leading to **adaptation** mechanisms to change iteratively by using insights from tracked actions and their outcomes (e.g., Bousdekis et al. 2020; Zhang et al. 2021),. This adaptation is driven by the implications of decisions

made within the decision environment and dynamic shifts in decision inputs, particularly in dynamic and evolving contexts. Also, this can be done autonomously by observing the decision-maker or the outcomes in the decision environment (Liu et al. 2019; Tamimi et al. 2019), but also with the decision-maker's input (e.g., Krumeich et al. 2016; Kim et al. 2020; Vater et al. 2020; Miikkulainen et al. 2021). Action mechanisms are crucial in distinguishing between the PAS archetypes we identified in our synthesis. Therefore, we will delve deeper into these aspects in the respective Sect. (4.2).

4.1.6 Ancillary features

In addition to the formulation, input, processing, and action mechanism components, our literature review has revealed further concepts within the context of PAS. We term these concepts "ancillary" since they are not at the core but somewhat secondary to PAS functionality. The literature encompasses many features, from data properties and general infrastructure considerations to integrating PAS within manufacturing systems (e.g., Vater et al. 2019; Ansari et al. 2019; Consilvio et al. 2019). Our focus, however, remains on overarching aspects and concepts that align with our research objectives. Specifically, we concentrate on features directly connecting to how PAS is situated within the organizational context and their role in decision-making processes.

Integration is a central concept determining a PAS's positioning in the broader (inter-)organizational landscape. Vertical integration allows data and decision outputs to be available across hierarchical levels, while horizontal integration incorporates data throughout processes or business functions, even externally (Appelbaum et al. 2017; Kumari and Kulkarni 2022). The growing data volume and sophisticated algorithms demand increased computing power, addressed through **distributed computing**, linking computational resources for shared data and processing power (Lepenioti et al. 2020). **Modularization** reduces complexity by, for instance, separating descriptive, predictive, and prescriptive analytics or specific functions (Appelbaum et al. 2017; Frazzetto et al. 2019). Additionally, **security- and privacy-preserving features**, though a niche in current research, are crucial due to rising cyber threats. For example, Harikumar et al. (2022) propose an algorithm for private prescription vectors.

With a focus on the decision-maker's perspective, our review has unearthed studies discussing **explainability** within PAS-based decision-making. The objective is to bolster user trust in the decision-making process, facilitating the adoption and effective implementation of system recommendations in real-world scenarios (e.g., Mehdiyev and Fettke 2020; Notz 2020; Suvarna et al. 2022). **Visualization** is pivotal as a design feature in PAS, guiding users visually through the decision process. Visualized results prove instrumental in enabling users to swiftly grasp decision outcomes and potential consequences (e.g., Appelbaum et al. 2017). The **workflow interface** serves as a guiding element, allowing users to navigate the decision process with the flexibility to adjust input parameters, underlying models, or output validation. These adjustments can be facilitated through no-code or traditional programming interfaces (e.g., Frazzetto et al.

2019). Furthermore, **extensibility** options are paramount in PAS, allowing users to install or develop components tailored to specific use cases (e.g., Frazzetto et al. 2019).

4.2 System archetypes

Per our study's objectives, this section delves into PAS from a meta-level perspective. The overarching goal is the enhancement of organizational decision-making. Building on the decision phases in the background section, we use these as the foundation to conceptualize archetypes, denoting overarching designs or setups with distinct characteristics. Through this lens, we aim to illuminate the synergy among the technical and the social subsystems, the prescriptive agent and human decision-maker, respectively, underscoring their collective role in refining organizational decision-making processes.

Given the extensive body of literature, configurations of the constituent components are diverse, depending on the industry, application, or specific use case. While many authors primarily emphasize technical aspects, the analysis of this literature reveals recurring patterns in the overall structure of PAS. The patterns are often anchored in the general decision-making phases (evaluation of alternatives, decision-making, and adaptation). One noteworthy observation is that technical subsystems are not uniform in their role within the decision-making process, nor their interaction with human decision-makers, and within the reviewed literature, a discernible shift emerges. Prescriptive analytics is evolving from a passive tool used by human decision-makers to having agency and assuming responsibilities of their own in the decision-making process. Prescriptive agents exhibit a growing decision-making latitude, and they can assume the role of substitutes for behavior-based or outcome-based decision-making by prescribing, autonomously executing actions, and adapting to changes in the decision environment (Baird and Maruping 2021).

To conceptualize these findings, we draw upon the theoretical framework of IS delegation proposed by Baird and Maruping (2021), anchored in agent interaction theories. Specifically, we adopt delegation mechanisms to delineate and explain four distinct system archetypes: *advisory*, *executive*, *adaptive*, and *self-governing* PAS. Our focus is directed toward understanding the (i) levels of delegation and the (2) roles or responsibilities played by human decision-makers and prescriptive agents across the three decision-making phases. Table 4 summarizes the key characteristics of each archetype, and a complete overview of the identified archetypes in our literature sample is available in Appendix E. Additionally, Table 5 demonstrates the roles and authority of the prescriptive agent and human decision-maker in each archetype. The agency of the other is not always entirely removed, and it rather pertains to the primary responsibility of a delegator or proxy-based relationship (Baird and Maruping 2021). We detail this in the following sections, where we will analyze the archetypes identified through our study, supported by exemplary visualization (refer to Figs. 7, 8, 9, and 10) of the responsibilities and delegation mechanisms in the PAS-based decision-making process.

Table 4 Summary of key characteristics of PAS archetypes

Archetypes	Key characteristics
Advisory PAS	• Prescriptive agents support the first phase of the decision-making process by prescribing an output to the decision-maker as a primarily passive tool to be used • The decision-maker takes this output and manually takes action in the decision environment • The prescriptive agents are disconnected from the decision environment and do not adapt to changes, which humans must perform manually
Executive PAS	• The prescriptive agent evaluates the alternatives and takes action in the decision environment • The prescriptive agent remains static and does not adapt to the effects of the outcomes on the environment • The human decision-maker remains responsible for observing the effects and adapting
Adaptive PAS	• The prescriptive agent prescribes an output to the decision-maker, who executes the decision in the environment • By observing the decision-maker or incorporating human feedback, the prescriptive agents adapt to the consequences and effects of the prescribed output in the environment
Self-governing PAS	• The archetype combines the capabilities of the three former archetypes • The prescriptive agent has agency and responsibilities in the whole decision-making process, mainly with some involvement from the human decision-maker but with the potential for full automation • There is a high potential for delegation between prescriptive agents and human decision-makers in the phases

4.2.1 Advisory PAS

The advisory archetype is notably the most common variant in the literature sample by a significant margin. In this archetype, prescriptive agents contribute only to the initial phase by assessing alternatives and presenting the optimal decision or course of action to the user, who maintains full decision-making authority. These prescriptive agents are static and do not adapt to the consequences of a decision or changing environments, necessitating manual adjustments or reconfigurations of inputs or underlying models by humans. The prescriptive agents are, in this sense, mostly passive tools to be used by the decision-maker with minimal delegation or agency in the decision-making process and entirely disconnected from the problem environment. For instance, Abdollahnejadbarough et al. (2020) explore a telecommunications provider employing an advisory PAS for supplier management. The system collects data from internal ERP and external supplier sources before employing machine learning to cluster suppliers. Subsequently, an optimization engine processes the results to recommend the most efficient suppliers for sourcing decisions. The decision-maker handles the following steps, such as contacting suppliers or executing purchase orders, without further involvement of the prescriptive agent.

Though less extensively researched, there are some examples in the literature where delegation does happen between the decision-maker and the prescriptive agent during the evaluation of alternatives phase (cf. Figure 7). This interaction might involve adjusting inputs or decision variables to accommodate real-world

Table 5 Overview of PAS archetypes and the respective primary authority of human decision-makers and prescriptive agents in the decision-making phases

PAS archetypes	Evaluation of alternatives	Decision-making	Adaptation and learning	Frequency in literature sample	Example articles
Advisory	Prescriptive agent	Human decision-maker	Human decision-maker	197	[29], [66], [122], [179], [244], [36], [64], [260]
Executive	Prescriptive agent	Prescriptive agent	Human decision-maker	12	[23], [98], [128], [163], [223], [199], [251]
Adaptive	Prescriptive agent	Human decision-maker	Prescriptive agent	21	[11], [198], [218], [220], [227], ([32], [79], [92], [162], [234]
Self-governing	Prescriptive agent	prescriptive agent	Prescriptive agent	13	[16], [24], [51], [67], [159], [21], [60], [77], [196], [210]

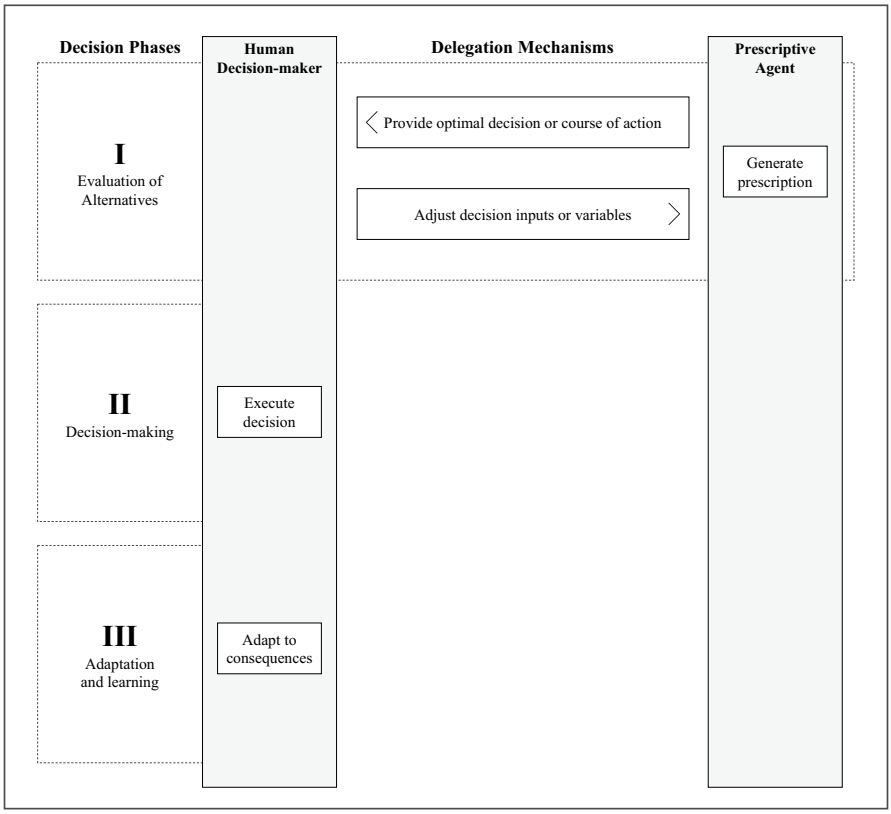


Fig. 7 Exemplary visualization of delegation mechanisms and responsibilities in advisory PAS

factors, expert knowledge, or risk preferences by the decision-maker. For example, Kawas et al. (2013) outline a PAS for sales team assignments that recommends optimal allocations while allowing decision-makers to fine-tune output through what-if analyses. This approach incorporates expert judgment (i.e., expert-in-the-loop), such as customer sentiment or subjective preferences, enabling experimentation with diverse sales team configurations.

4.2.2 Executive PAS

Executive PAS is the least common archetype in our sample. With only twelve papers, this archetype represents a niche in current PAS research. Humans traditionally hold the mandate to act upon a prescriptive output. However, the literature also suggests some PAS designs in which the prescriptive agent receives the authority to execute decisions autonomously in the problem environment. In these systems, adaptation remains static, and the prescriptive agent is responsible for the initial two phases, with minimal interaction with the decision-maker.

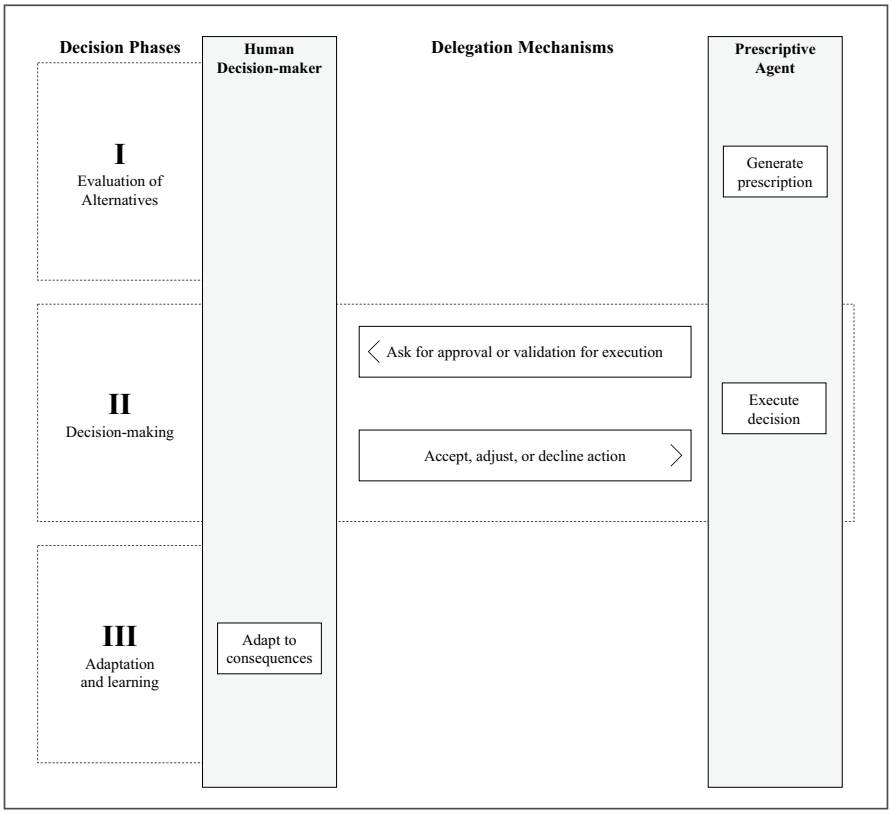


Fig. 8 Exemplary visualization of delegation mechanisms and responsibilities in executive PAS

Executive PAS are predominantly utilized in domains with high automation, standardization, or repetitiveness, where rapid decision-making is necessary.

For example, Soroush et al. (2020) introduce a PAS recommending optimal safety actions to detect and address process operation hazards by implementing mitigative chemical-process measures. Similarly, Mazon-Olivo et al. (2018) describe a PAS in precision agriculture that autonomously sends repetitive and planned actions to IoT devices in the field. Additional examples include intelligent call center routings where an optimal service employee is matched to a customer (Ali 2011) and a data allocation scheme across a Hadoop cluster for enhanced data security and privacy (Revathy and Mukesh 2020). In some cases (cf. Figure 8), there is a higher degree of delegation, where the prescriptive agent executes the decision in the environment, but a decision-maker must first approve or validate the output (Rizzo et al. 2020). This approach can benefit high-stakes decision-making with significant financial implications or safety and compliance concerns.

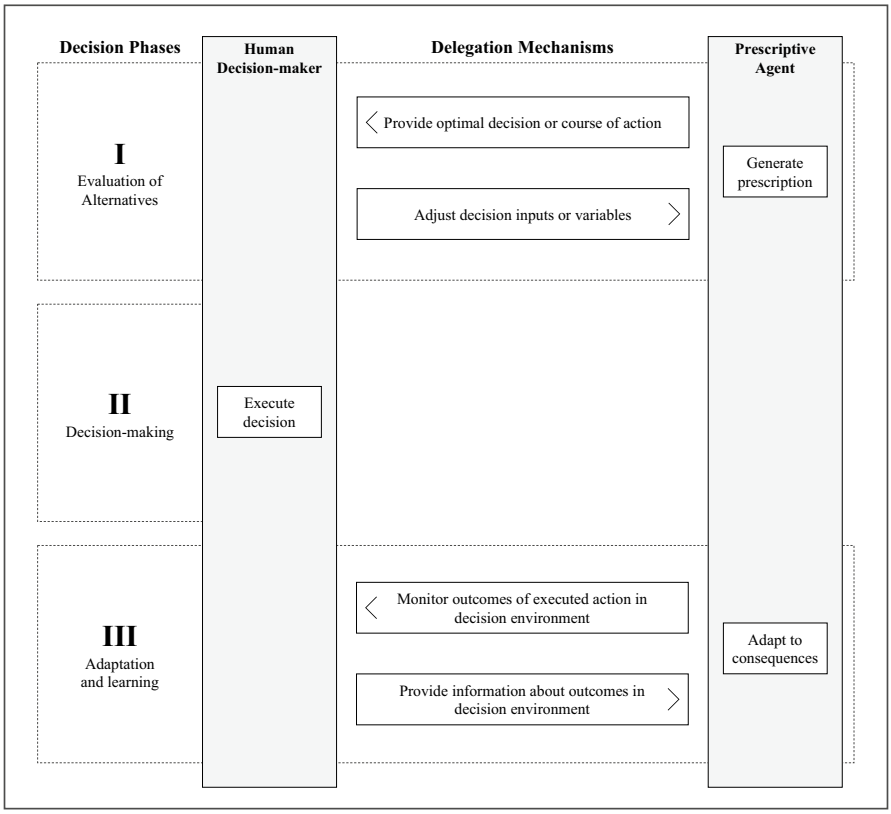


Fig. 9 Exemplary visualization of delegation mechanisms and responsibilities in adaptive PAS

4.2.3 Adaptive PAS

In the case of an Adaptive PAS, while human decision-makers maintain authority over the decision-making phase, the prescriptive agent assists in the adaptation and learning phases, contributing to a more effective and well-informed decision-making process. The prescriptive agent monitors decision outcomes and their impact on the decision environment, incorporating observations into subsequent iterations by adding new data as input or dynamically adjusting the decision model. As problem environments often change due to shifting requirements, priorities, or new knowledge, adaptive PAS, as a dynamic archetype, holds significant potential compared to static counterparts.

For example, Liu et al. (2019) propose a system for optimizing locomotive wheel maintenance operations, recommending inspection schedules to minimize long-term cost rates. Similarly, Zhang et al. (2021) present a reinforcement learning-based maintenance optimization model that determines optimal actions based on a machine's ever-changing degradation state. Bousdekis et al. (2020) emphasize the importance of feedback and learning mechanisms in a generic

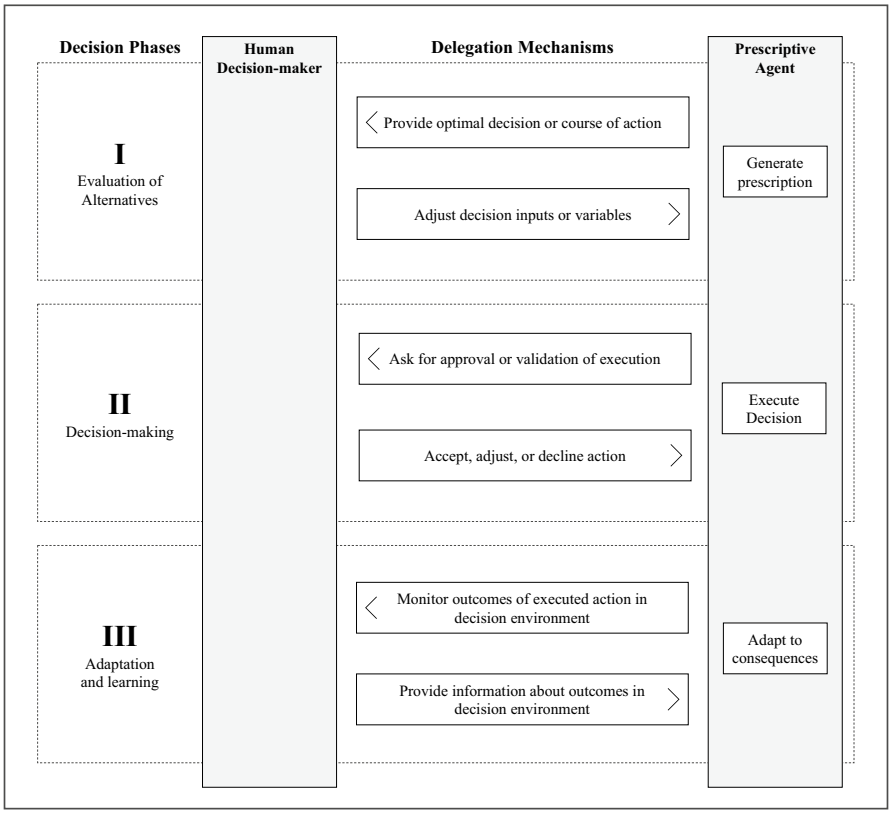


Fig. 10 Exemplary visualization of delegation mechanisms and responsibilities in self-governing PAS

IoT scenario, where an agent suggests optimal actions to users and updates the prescriptive model dynamically based on real-time IoT sensor data. Prescriptive Agents can also observe the decision environment and decision-makers while actioning. Tamimi et al. (2019) discuss a PAS for field development design, recommending optimal designs and deriving the decision-maker's utility function for subsequent iterations of prediction and optimization models. Käki et al. (2019) highlight the deviation from model recommendations in production planning, often resulting in deteriorated performance, and emphasize the added value of the adaptive PAS compared to static counterparts.

There are also examples in the literature where decision-makers observe action consequences or judge potential outcomes based on domain knowledge (i.e., expert-in-the-loop systems). Here, the prescriptive agent requires active input from the human to adapt for subsequent iterations, with decision-makers providing information by, for example, relabeling outputs or aggregating real-world outcomes into the training set for future cycles (e.g., Krumeich et al. 2016; Kim et al. 2020; Vater et al. 2020; Miikkulainen et al. 2021). This archetype indicates that expert knowledge and human judgment remain vital in the adaptation and

learning phase. Also, from the perspective of GST, adaptation is crucial, as the social and technical subsystems naturally change over time (Chatterjee et al. 2021).

4.2.4 Self-governing PAS

The fourth archetype, self-governing PAS, represents a potentially fully autonomous system where the prescriptive agent has agency and responsibilities in the entire decision process independently or with the decision-maker's involvement. Combining the capabilities of the other three archetypes, the self-governing PAS is the most sophisticated version, offering the highest added business value due to automated execution, adaptability, dynamic self-learning mechanisms, reduced manual work, and enabling rapid, fact-based decision-making in dynamic environments.

Self-governing PAS relates to well-researched and practiced areas such as route optimization and data load distribution (cf., Wang et al. 2008; Jozefowiez et al. 2008), used in highly structured environments, which could be considered precursors or early manifestations of the archetype. However, they differ from more recent examples in aspects like their integration into the broader infrastructural landscape and their use of historical and real-time data. Moreover, being less disjointed from the decision environment, they support the entire decision-making process while showing a high potential for delegation between the human decision-maker and the prescriptive agent.

Examples from our literature sample primarily come from domains with high technological maturity and sophistication, such as cyber-physical systems, IoT, modern energy distribution systems, or smart-sensor-driven environments. These technologies are inherently data-driven, automated, and integrated—ideal preconditions for advanced agents. For instance, Ceselli et al. (2019) propose a data-driven framework for optimally distributing data traffic from mobile access points across capacity-constrained mobile edge cloud networks. A fully autonomous orchestrator module executes the best data assignment plans. The selected plans, their effects, and the access points' demands are continuously logged and validated for subsequent iterations. Similarly, Vater et al. (2020) introduce an IoT-based architecture for real-time error detection in automotive manufacturing, utilizing edge-/cloud-architecture including modules for preprocessing, prediction, prescription, action-taking, and validation to close the loop for a fully autonomous decision-making process.

Gutierrez-Franco et al. (2021) present a PAS for last-mile delivery operations as another example. The system leverages historical data such as traffic, customer behavior, and driver performance as input. This data is initially preprocessed and descriptively analyzed, forming a foundation for predicting future operations and prescribing optimal routes or schedules. The generated output is fed into an execution module, providing optimal routes for drivers. Furthermore, real-time circumstances, including traffic or route deviations by drivers, are captured via GPS or the vehicle's sensors, allowing continuous recalculation of the optimal schedule. In this instance, the decision process is not linear but dynamically adapts based on the current state of the problem environment. At the end of each shift, a learning

mechanism initiates, collecting accumulated data and best practices to enhance delivery operations, serving as historical data input for subsequent days.

4.3 Technology affordances

As outlined in the scope of our SLR, our goal is to uncover affordances that represent specific purposes or decision-making tasks supported by PAS. The resulting concept matrix begins with industry (bold text) and is divided into specific affordances (italic text). We use affordances in the sense of action potential. The technical nature of our literature sample poses a challenge in extracting both the perception and actualization of affordance. In this context, we can only derive how the social subsystem perceives the affordance, for example, through visualization. Conversely, actualization can only be detailed based on the actor responsible for executing the decision, either the technical component or the human decision-maker (Pozzi et al. 2014; Leidner et al. 2018) (cf. Figure 11 for details), which we detail in the previous section on system archetypes.

Further, our literature sample includes papers that do not address a specific affordance but provide a more general perspective, such as mathematical or algorithmic formulations, infrastructural considerations, reviews, and conceptual papers. Therefore, they are excluded from our considerations in this section. In the following, we will focus on more prevalent industries (N > 5) and affordances, referring to Table 6 for niche examples. We conclude this section with a summary and overarching affordance patterns. Additionally, Appendix D provides a comprehensive overview of all affordances mapped to the literature sample. Further, Appendix H includes a trend analysis across the more prominent industries.

4.3.1 Manufacturing

As the most researched industry, manufacturing has historically relied on mathematical models to optimize processes, workforce allocation, and schedules, which are

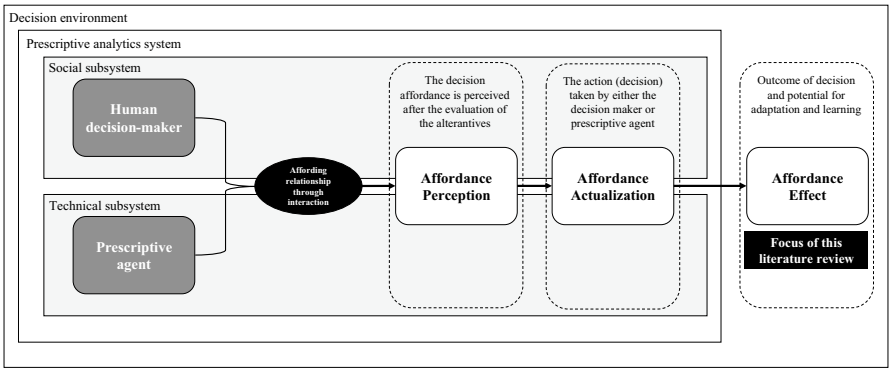


Fig. 11 Affordance theoretical framework mapped to PAS and the decision-making phases (own depiction based on Pozzi et al. 2014)

Table 6 Concept matrix of PAS-driven decision-making affordances

Industry; frequency	Affordance (effect); frequency	Example studies (IDs)
Manufacturing; n = 44	Maintenance planning for optimal maintenance schedule; n = 24	[11], [54], [137], [196]
	Production planning for optimal manufacturing schedule; n = 13	[43], [61], [123], [226]
	Product (portfolio) design optimization; n = 4	[66], [202], [213], [216]
	Operations safety improvement and planning; n = 1	[128]
	Industrial worker training optimization; n = 1	[91]
	Deformation control in additive manufacturing; n = 1	[174]
Transportation and logistics; n = 25	Optimization of routing and scheduling; n = 15	[19], [67], [131], [237]
	Capacity/cargo management and improvement; n = 8	[53], [88], [173], [184]
	Vehicle maintenance planning; n = 2	[10], [200]
Health and MedTech; n = 26	Patient treatment planning and improvement; n = 8	[39], [99], [142], [172]
	Patient scheduling; n = 8	[8], [130], [146], [197]
	Pandemic/epidemic intervention planning; n = 5	[6], [83], [92], [122]
	Human health tracking and improvement; n = 3	[1], [71], [168]
	Assortment and inventory planning; n = 1	[158]
	Clinical investment management; n = 1	[15]
Energy and environment; n = 24	Optimizing power system/grid operations; n = 5	[7], [25], [31], [262]
	Disaster preparation/recovery planning; n = 3	[117], [138], [214]
	Electricity brokerage optimization; n = 3	[24], [205], [206]
	Maintenance planning; n = 3	[27], [45], [201]
	Wastewater treatment improvement; n = 2	[18], [253]
	Waste collection and management planning; n = 2	[107], [114]
	Optimization of deepwater casing exits; n = 2	[186], [195]
	Waterflooding process optimization; n = 1	[103]
	Battery lifetime optimization; n = 1	[218]

Table 6 (continued)

Industry; frequency	Affordance (effect); frequency	Example studies (IDs)
	Reservoir design planning; n = 1	[32]
	Soil slope analysis; n = 1	[119]
Retail and trade; n = 12	Price optimization; n = 4	[46], [76], [135], [209]
	Assortment and inventory planning; n = 3	[94], [110], [152]
	Sales team assignments; n = 2	[36], [161]
	Customer characterization; n = 1	[52]
	Customer service recommendation; n = 1	[96]
	Theft surveillance and automated checkout; n = 1	[51]
Education; n = 10	Academic performance improving; n = 6	[34], [50], [109], [245]
	Dropout prevention planning; n = 2	[113], [212]
	Admissions planning and selection; n = 2	[59], [252]
Chemicals and resources; n = 8	Maximize oil/gas recovery; n = 4	[85], [143], [156], [163]
	Mining fleet scheduling; n = 1	[33]
	Sand molding process improvement; n = 1	[63]
	Laboratory task allocation and planning; n = 1	[35]
	Biodiesel properties optimization; n = 1	[147]
Technology and communications; n = 6	Social media usage optimization; n = 2	[234], [238]
	Network and computing resource orchestration; n = 2	[136], [169]
	Software development estimation; n = 1	[235]
	Website performance analysis and optimization; n = 1	[81]
Academia; n = 6	Research advising; n = 6	[170], [179], [180], [219]

Table 6 (continued)

Industry; frequency	Affordance (effect); frequency	Example studies (IDs)
Agriculture; n = 5	Harvesting operations planning and optimization, n = 3	[90], [105], [223]
	Crop yield optimization; n = 1	[256]
	Fish trawling routing and optimization; n = 1	[221]
Crime and law enforcement, n = 3	Law enforcement resource allocation and planning; n = 2	[111], [187]
	Imprisonment decision planning and recommendation; n = 1	[248]
Sports and recreation; n = 3	Tournament lodging planning; n = 1	[164]
	Sports event safety management and planning; n = 1	[224]
	Athlete training process improvement; n = 1	[124]
Defense and military; n = 2	Infantry engagement planning; n = 1	[118]
	Military logistics planning; n = 1	[261]
E-Commerce; n = 2	Markdown planning and price optimization; n = 1	[28]
	Oder delivery scheduling; n = 1	[14]
Finance; n = 2	Teller machine replenishment planning and allocation; n = 1	[217]
	Stock purchase recommendations; n = 1	[241]
Media; n = 2	Radio advertising scheduling; n = 1	[126]
	Movie planning and profit-maximizing; n = 1	[78]
Professional services; n = 2	Project staffing planning and allocation; n = 2	[167], [228]
Public sector; n = 2	Child welfare assessment; n = 1	[153]
	Infrastructure planning and optimization; n = 1	[178]
Tourism & hospitalitiy; n = 2	Optimal tour pricing; n = 1	[127]
	Restaurant recommendations; n = 1	[57]

Table 6 (continued)

Industry; frequency	Affordance (effect); frequency	Example studies (IDs)
Industry-agnostic; n = 20	Prescriptive process management; n = 8	[40], [95], [192], [203]
	Employee recruiting and staffing; n = 3	[13], [80], [160]
	Procurement and supplier management; n = 2	[177], [258]
	Marketing management; n = 2	[70], [125]
	Facility/asset management; n = 2	[116], [129]
	Managerial Accounting; n = 1	[104]
	Call center routing; n = 1	[112]
	Server incident management and prevention; n = 1	[132]

crucial in enhancing efficiency and productivity. With Industry 4.0, manufacturing has transformed into a highly developed sector integrating advanced technologies like robotics and IIoT to establish intelligent, interconnected production systems (Wanner et al. 2023). As reflected in our literature sample, these developments have spurred extensive research interest in data-driven analytics. *Maintenance planning*, a dominant affordance for PAS, has emerged as a research stream called prescriptive maintenance. Here, the primary purpose is to afford optimal maintenance schedules, often incorporating spare parts management, primarily driven by machine sensor data. For example, these systems employ descriptive analytics to analyze the current machine state and predictive analytics to foresee potential failures, fueling a prescriptive model to propose the optimal schedule (e.g., Liu et al. 2019; Ansari et al. 2019; Fox et al. 2022). Similarly, *production planning*, a longstanding research area, has begun to harness sensor-driven data to improve production schedules, processes, quality, and operations. Examples include PAS to optimize shop floor operations (Stein et al. 2018) and schedule diffusion furnaces (Vimala Rani and Mathirajan 2021).

In addition to the two dominant affordances, there are other, less explored examples within the manufacturing sector. Some of these include supporting *product development*, optimizing product portfolio designs (Jank et al. 2019), and enhancing the design of industrial products (Dey et al. 2019). Moreover, research has proposed utilizing PAS for *training industrial workers* by offering training schedules based on digital twins (Longo et al. 2023), prescribing *optimal safety actions* (Soroush et al. 2020), and *improving additive manufacturing processes* through deformation control (Jin et al. 2016). As the manufacturing industry continues to advance, the potential applications of PAS are expected to grow, fostering further innovation and efficiency.

4.3.2 Transportation and logistics

Facing significant pressure regarding cost efficiency and sustainability, the transportation and logistics industry, like manufacturing, has a history of using mathematical optimization models for routing and scheduling tasks (Konstantakopoulos et al. 2022). With vehicles becoming increasingly connected and generating digital traces through sensors and networks, PAS can further harness this data to afford efficiency, enabling the industry to capitalize on various applications. *Routing and scheduling* naturally emerge as the most researched affordance.

Examples span various modes of transportation, such as ground (Gutierrez-Franco et al. 2021), air (Ayhan et al. 2018), and public transport (Xylia et al. 2016), showcasing the versatility and potential of PAS in enhancing efficiency across diverse systems. Another affordance is *capacity management*. Affordance effects include optimizing freight or cargo distribution (Rizzo et al. 2020) and passenger seat assignments (Moore et al. 2021), ensuring efficient resource allocation, and enhancing overall operational performance. Lastly, considering that vehicles undergo continuous degradation while in use, our sample also includes prescriptive maintenance affordances to effectively address wear and tear,

optimize maintenance schedules, and prolong the service life of vehicles (Consilvio et al. 2019; Anglou et al. 2021).

4.3.3 Health and MedTech

Health and MedTech are the second most researched sectors in our literature sample. Implementing PAS can significantly improve resource allocation and overall patient outcomes with the increasing complexity and demand for healthcare services. The sector features two dominant affordances: patient treatment planning and scheduling. *Patient treatment planning* primarily focuses on improving health outcomes by optimizing treatments, reducing hospital readmissions, enhancing precision medicine, and boosting clinical staff efficiency (Rider et al. 2021; Zheng et al. 2021). On the other hand, *patient scheduling* emphasizes the efficient management of appointment scheduling and bed occupancy (Belciug and Gorunescu 2016; Srinivas and Ravindran 2018). In the clinical context, niche examples of affordances include *assortment, inventory planning* (Galli et al. 2021), and *investment management* (Fang et al. 2021).

Further, in light of the recent COVID-19 pandemic, several contributions have focused on *pandemic or epidemic intervention planning*. These studies consider various aspects, such as mobility intervention and the rapid deployment of medical staff and equipment (Miikkulainen et al. 2021; Ahmed et al. 2021). Lastly, a group of researchers has shifted their focus to the patients or their bodies directly, for example, incorporating sensor data for *health tracking*, enhancing safety and consumption decisions, and preventing impulsive behavior among patients (Sedighi Maman et al. 2020; Raychaudhuri et al. 2021). By leveraging PAS in these areas, healthcare providers can offer more personalized care, empower patients to make better-informed decisions, and ultimately improve overall health outcomes.

4.3.4 Energy and environment

The energy and environment industry showcases more diverse affordances in our literature sample than in previous sectors. A key focus in this domain is power generation systems (e.g., wind farms), where PAS are utilized to afford *optimized performance* (Tektaş et al. 2022), *electricity brokerage* (Peters et al. 2013), and *prescriptive maintenance* (Goyal et al. 2016) for these systems. Additionally, PAS applications extend to *optimizing waste collection and planning* (Vargas et al. 2022) and enhancing *wastewater treatment processes* (Zadorojniy et al. 2019). Some PAS afford *disaster preparation and recovery planning*, addressing challenges posed by wildfires, hurricanes, or floods (Hu et al. 2019; Yang et al. 2022). Niche examples within this sector include *soil slope analysis* (Li et al. 2019) and *optimization of battery lifetime* (Eider and Berl 2020). As demonstrated for this sector, implementing AI systems can significantly impact society by improving sustainability, efficiency, and environmental protection (Schoormann et al. 2023).

4.3.5 Retail and trade

The retail and trade industry encompasses both B2B and B2C interactions. Despite its size, there has been relatively little PAS research in this area compared to other industries. One possible reason may be the traditional set-up often found in brick-and-mortar stores and a comparatively lower level of digitization than in industries like manufacturing or logistics. Nevertheless, there are instances of PAS research in our sample. One example is dynamic *price optimization*, which incorporates factors such as customers, competition, business partners, and environmental aspects (Ito and Fujimaki 2017). A key challenge in this industry is *assortment and inventory planning*, which is a significant cost driver when considering perishable goods or inventory costs (Jin et al. 2016; Flamand et al. 2018). In the B2B context, *sales teams* drive revenue, making optimal assignment a potential affordance. Other examples include *customer characterization* (Perugini and Perugini 2014), *customer service recommendations* (Lo and Pachamanova 2015), *theft surveillance, and facilitating automated store checkouts* (Hauser et al. 2021).

4.3.6 Education

Three primary affordances have emerged in education: dropout prevention planning, improving students' academic performance, and admissions planning. Firstly, *dropout prevention planning* focuses on identifying students at risk of leaving their educational programs prematurely. PAS enables institutions to target support and interventions, ensuring that students receive the help they need to stay on track and complete their studies (Yanta et al. 2021; de Jesus and Ledda 2021). Secondly, *improving academic performance* is an essential priority for educational institutions. By utilizing PAS, educators can gain insights into students' learning patterns and areas of difficulty, enabling them to tailor teaching approaches and offer personalized learning experiences that foster success (Uskov et al. 2019; Islam et al. 2021). Finally, *admissions planning* is essential to maintaining a thriving educational institution. PAS can help optimize the admissions process by analyzing student demographics and academic performance to ensure that institutions admit the most suitable candidates (Kiaghadi and Hoseinpour 2023). Applying PAS in education can improve traditional academic decision-making processes, enhance student outcomes, and streamline institutional operations.

4.3.7 Chemicals and resources

The chemicals and resources industry has some affordances identified in our review, aiming to optimize processes, boost efficiency, and promote sustainable resource utilization. *Maximizing oil and gas recovery* is the most researched affordance effect, with PAS used to optimize extraction techniques, reservoir modeling, and resource management. Other examples include *laboratory task allocation* (Silva and Cortez 2022), *mining fleet scheduling* (Nakousi et al. 2018), *optimizing biodiesel properties* (Suvarna et al. 2022), *and improving sand molding processes* (Chowdhary and Khandelwal 2018).

4.3.8 Technology and communication

Despite its technological maturity, the technology and communications sector has limited research on PAS applications. This sector can benefit from PAS in various affordances, such as *network and computing resource orchestration* (Ceselli et al. 2018, 2019), *social media optimization* (Ballings et al. 2016), *software development estimation* (Pospieszny 2017), and *website performance analysis* (Salvio and Palaoag 2019). PAS can streamline network and computing systems, enhance social media campaigns, improve project management, and optimize website performance. Although current literature is limited, this industry has the potential for further exploration. Harnessing PAS can improve performance, efficiency, and user satisfaction across technology and communication operations.

4.3.9 Academia

Furthermore, several studies have investigated the use of PAS in academia. They focus on improving and enhancing academic research performance. By analyzing data related to research output and other relevant factors, such as citations or related work, PAS can provide valuable insights and recommendations for researchers, such as journals or references. Additionally, some research has explored the concept of system thinking and crafting scenarios. These approaches help researchers better understand their work's potential outcomes and consequences, enabling them to make more informed decisions about their research directions (e.g., Song et al. 2014; Jeong and Joo 2019).

4.3.10 Industry-agnostic

Beyond sector-specific applications, much research investigates affordances from an industry-agnostic perspective. In these cases, authors often apply prescriptive analytics in a context, which we coin *prescriptive process management* (Krumeich et al. 2016; Kubrak et al. 2022). The primary affordance effects include process monitoring, controlling execution, and recommending the most appropriate subsequent actions. Interestingly, two contributions specifically discuss the explainability of decision outputs in this setting (Mehdiyev and Fettke 2020; Notz 2020). Moreover, PAS can be used to optimize *employee recruitment* (Pessach et al. 2020), *facility and asset management* (Lavy et al. 2014), and *supplier selection* (Abdollahnejadbarough et al. 2020). Some niche systems involve enhancing *accounting* procedures and improving *call center routing* (Ali 2011). Overall, these versatile applications demonstrate the potential of PAS to streamline operations and support decision-making across a wide range of industry contexts.

4.3.11 Summary

In conclusion, PAS have demonstrated the potential to affect various industries positively, for instance, by optimizing processes, allocating resources, scheduling, and planning maintenance actions. The systems afford organizations to achieve enhanced efficiency and productivity. Although the current research varies in scope and depth

across different sectors, the widespread applicability of PAS indicates its capacity to drive innovation and streamline operations across a wide range of contexts. It is important to note that this section did not detail all affordances, specifically under-represented industries such as tourism, media, and finance. Please refer to Table 6 and the example papers for more information on these sectors and their respective affordances.

Even though the PAS in our literature sample are diverse, they share the commonality of prescribing the best course of action in a specific decision environment and situation. Given the abovementioned affordances, we derived three overarching affordance effects of PAS to improve decision-making processes: (1) improvement, (2) scheduling, and (3) resource allocation, which we detail in the following.

Improvement (1) is centered on enhancing and optimizing the current state of an object or the decision environment to achieve an improved state. This affordance involves conducting a comprehensive analysis and adjusting processes, products, or operations to make them more effective, efficient, and aligned with predefined goals. For example, product design optimization focuses on continuously refining products based on user feedback and market trends (e.g., Dey et al. 2019; Jank et al. 2019). Likewise, within the healthcare sector, patient treatment planning and improvement entail tailoring treatments to individual patient needs while constantly updating these plans based on patient responses and emerging medical insights (e.g., Rider et al. 2021; Zheng et al. 2021). Similarly, maintenance optimization aims to optimize equipment performance and longevity in industrial settings, ultimately reducing downtime and increasing efficiency (e.g., Liu et al. 2019; Consilvio et al. 2019).

Scheduling (2) entails strategically organizing and coordinating tasks and actions in the decision environment over time, creating and managing a timeline of activities that aligns with an organization's objectives, resource availability, and external factors. Effective scheduling and planning improve operational flow and resource utilization, reducing bottlenecks and inefficiencies. For example, in healthcare, patient scheduling is critical for maximizing medical facilities and staff use (e.g., Belciug and Gorunescu 2016; Srinivas and Ravindran 2018), while in the energy sector, planning energy distribution is crucial for balancing supply with consumer demand (Goyal et al. 2016).

Resource allocation (3) involves strategically distributing resources like workforce, materials, and finances to areas most needed and will be most effective. This process requires a thorough understanding of resources' availability, potential, limitations, and different organizations' objectives and needs. For example, in logistics, capacity and cargo management ensures optimal use of transport and storage resources (Rizzo et al. 2020; Gutierrez-Franco et al. 2021). In urban planning, efficient allocation of resources for waste collection and management is vital for maintaining cleanliness and public health (Vargas et al. 2022).

The affordances are not mutually exclusive but can overlap depending on the specific system, application use case, or complex organizational settings. For instance, in manufacturing, the improvement of a product design (Improvement) is closely linked to the planning of production schedules (Scheduling) and the allocation of manufacturing resources (Resource Allocation). Similarly, in healthcare, patient treatment plans (Improvement) need to be integrated with patient scheduling

(Scheduling and Planning) and the allocation of medical staff and equipment (Resource Allocation). Understanding the interplay and overlap of these affordances is crucial for effective management and decision-making. Organizations can holistically approach problem-solving and optimization by recognizing how they complement each other, leading to more comprehensive PAS.

5 Directions for future research

Drawing on the synthesis and conceptualization in the previous sections, we discuss the main observations and deriving aspects that remain open to pave the ground for future research. We will discuss possible research directions from technical, social, and overarching perspectives. Table 7 highlights the research agenda, key observations, and illustrative research directions or questions.

5.1 Technical perspectives

The preponderance of research within the technical subsystem is understandable, as PAS are fundamentally rooted in technology, and a significant portion of investigations in this area stem from disciplines closely tied to technological advancements. Our SLR has uncovered numerous PAS components that various authors in our sample have extensively researched and well-addressed, establishing a solid foundation in the field. Despite this wealth of knowledge, we have identified specific gaps that warrant further investigation.

For instance, action mechanisms have been relatively underexplored in the existing literature. We contend that these components are crucial for designing an effective PAS, as they drive the interaction between the prescriptive agent and the human decision-maker. Further, ancillary features with a focus on seamlessly integrating technology components within the social structure and the broader organizational landscape are lacking in current research.

In addition to the identified gaps, mathematical programming and, to a lesser extent, bio-inspired optimization algorithms have been well-established in prescriptive analytics. Recently, there has been a surge of interest in incorporating ML techniques to integrate predictions or probabilities as precursors to subsequent optimization processes. ML, particularly deep learning, is widely researched for managing extensive and high-dimensional datasets (Janiesch et al. 2021). However, challenges surrounding interpretability and explainability have hindered its adoption among decision-makers. While explainable and interpretable ML offer promising solutions (Zschech et al. 2022; Herm et al. 2022; Wanner et al. 2022), integrating these approaches into a PAS remains mainly open. Today, ML predominantly contributes to descriptive and predictive analytics, but increased transparency and trust, which are vital, especially for high-stakes decision-making, remain open. However, interpretable ML would be a valuable extension to existing prescriptive analytics approaches (Shollo et al. 2022).

Table 7 Research agenda with six key observations

Perspectives	Key observations	Illustrative research directions or questions
Technological Perspectives	(1) Action mechanisms in prescriptive analytics are underresearched	• Research on the design requirements of action, tracking, and adaptation mechanisms in PAS • Research on the design requirements of workflow interface for human-AI interaction
	(2) Interpretable ML and RL are possible as extensions to optimization models	• Interpretable ML as an extension to PAS and its unique requirements • RL and its unique characteristics, concepts, and requirements within the PAS framework
Social perspectives	(3) The human perspective is underrepresented in PAS-based decision-making	• Research on how to increase the trust of the decision-maker regarding the decision output of the technology components • Do explainability, accountability, fairness, and bias require PAS-specific research, or is it generalizable through AI/ML research?
	(4) Links to and the embedment in the broader organizational landscape or structure are missing	• How can a PAS effectively contribute to driving fact-based decision-making in organizations? • How can a PAS be embedded across different organizational contexts and structures?
General Perspectives	(5) Executive, adaptive, and self-governing archetypes are underexplored, as well as the human-agent interface that facilitates the interaction	• Research on case-agnostic blueprints with corresponding design principles and options for the archetypes • What interface forms can facilitate and improve the interaction between humans and agents in a PAS? • How can a PAS perceive and actualize affordances?
	(6) The review identified diverse affordances in the sense of action or action possibilities, but research along more nuanced affordance theories could help design more effective PAS	• Affordance-actualization: investigate how decision outputs from a PAS are implemented within organizations and the effects on decision-making processes • Affordance-network approach: examine how organizations achieve more significant outcomes by connecting a series of more immediate, concrete decision outcomes • Affordance-trajectories: the trajectory along which decision outputs from PAS travel in organizations

Further, RL has also garnered attention for its potential in PAS owing to its dynamic, adaptive, and iterative nature and its aptitude for addressing well-formalized decision-making problems (Greene et al. 2022). Researchers have already demonstrated the effectiveness of RL in the context of adaptive and self-governing PAS archetypes. However, our understanding is limited by the scarcity of literature on this topic. Consequently, further research is needed to explore RL's unique characteristics, concepts, and requirements within the PAS framework, ultimately contributing to a more robust understanding of its potential applications and benefits and an alternative to more traditional optimization techniques such as linear programming.

Finally, As a recent innovation in AI, foundation models – particularly large language models – have catalyzed a significant paradigm shift in the development of AI systems. This transformation has already profoundly impacted existing IT services and ecosystems while simultaneously enabling the creation of novel applications (Feuerriegel et al. 2024; Schneider et al. 2024). Through our review, we have observed that the application of foundation models in prescriptive analytics remains unexplored. However, we posit that this domain holds substantial potential. Future research should thus investigate leveraging the advanced capabilities of foundation models to enhance intelligent decision-making systems.

5.2 Social perspectives

As previously mentioned, the social subsystem (including the human decision-maker) has been relatively underresearched, which is somewhat understandable considering the technological origins of prescriptive analytics. However, we argue that increased attention to this perspective is essential for a more comprehensive understanding of the field, especially for the IS community.

Current research in this area often takes a case-specific approach, with insufficient consideration of the broader organizational landscape and how systems integrate into the larger picture. The outputs generated by these decision-making processes can have varying contextual implications, depending on factors such as the business unit or hierarchical level, ultimately influencing strategic and operational decision-making (e.g., Appelbaum et al. 2017). Consequently, we posit that a PAS will be most valuable if it is organization-wide, encompassing all decision processes and business functions, avoiding siloed structures, and made available (e.g., as-a-service) to all decision-makers tailored to their unique environments. From a technological standpoint, achieving this vision requires infrastructural features such as standardized data integration, interoperability, distributed computing, and effective API design (Lepenioti et al. 2020; Vieira et al. 2020; Verbraeken et al. 2021). However, further research is needed to identify the specific requirements unique to a PAS. Additionally, given that analytics is not solely a technology-driven concept and necessitates a cultural shift towards evidence-based decision-making, it is worth exploring whether a PAS should be a mere result of this shift or function as a driver or initiative to move organizations toward more fact-based decision processes.

Beyond the organizational setting, the human decision-maker's perspective warrants further research. Specifically, ML for predictive analytics has been extensively

studied regarding explainability, interpretability, accountability, fairness, and bias (e.g., Meske et al. 2022; Nadeem et al. 2022; Kraus et al. 2023). Due to the inherent differences between various analytics methods and outputs, particularly those focused on prescriptive analytics, we argue that additional research is needed, specifically in the context of PAS, such as the decision-maker's trust in decision outputs (Caro and de Tejada Cuenca 2023).

This line of inquiry will facilitate a deeper understanding of the challenges and opportunities associated with PAS. It will also foster their responsible development and deployment within organizations, ensuring that they align with ethical standards and contribute positively to decision-making processes.

5.3 General perspectives

Our synthesis identified four archetypes within PAS. Despite this progress, most PAS in our sample are predominantly advisory, while the executive, adaptive, and self-governing archetypes remain underexplored. This imbalance suggests a significant disconnect between problem environments and PAS, as these systems are often disjointed from the last two phases of the decision-making process. In real-world scenarios, many environments are constantly in flux due to natural changes or actions. This dynamic nature is frequently overlooked in current PAS research. However, action execution, adaptation, and learning mechanisms hold great potential, as they can help reduce information loss across iterations and improve decision-making processes over time while minimizing reliance on subjective or judgmental human experiences (Sturm et al. 2021). The BA community must address this gap by understanding the requirements of such systems and developing case-agnostic blueprints with corresponding design principles and options.

Furthermore, delegation mechanisms warrant increased attention, representing the initial steps toward hybrid intelligence systems and the symbiosis between agents and humans in decision-making (Dellermann et al. 2019; Peng et al. 2022). By focusing on these underexplored areas, researchers can contribute to a more comprehensive understanding of PAS, ultimately fostering the development of systems that effectively integrate advanced technology and human expertise in organizational decision-making processes.

In our review, we utilized the affordance theory to examine how individuals or organizations with specific objectives can leverage technology at a basic level. Given the nature of the papers in our sample, we focused on affordance effects. We argue that understanding the balance between technology's enabling and constraining aspects is crucial for designing effective PAS for organizational decision-making. Affordance theory offers a promising perspective for future research in analytics-driven organizational decision-making. One area to explore is the process of affordance-actualization, which involves understanding how the potential benefits of technology are actualized in the form of organizational outcomes (Strong et al. 2014). In the context of our work, this would involve investigating how decision outputs from a PAS are implemented within organizations and the effects on decision-making processes themselves.

Another direction is applying the affordance-network approach, which examines how organizations achieve more significant outcomes by connecting a series of more immediate, concrete decision outcomes within a network of interrelated affordances (Burton-Jones and Volkoff 2017). Furthermore, exploring the trajectory along which affordances travel can provide valuable insights into the processes and conditions that shape the perception and actualization of affordances in organizational decision-making (Thapa and Sein 2018). This line of inquiry can help elucidate how the potential benefits of a PAS are transformed into tangible outcomes in practice. By considering these aspects of affordance theory, researchers and practitioners could develop more effective and comprehensive PAS for decision-making in organizational settings.

6 Concluding remarks

Our research contributes to the knowledge of BA, specifically PAS, as the most sophisticated maturity level to support organizational decision-making with the highest potential business value. In this context, we conducted an SLR on the state of PAS research, emphasizing the IS artifact and GST as a theoretical framework. We reviewed 262 relevant contributions, enabling a holistic view of the field and its relevance to the broader BA community in research and practice, with three main contributions.

Our first contribution is the development of a concept matrix comprising 23 distinct constituent components, revealing fundamental technical design elements. Based on GST and considering important ancillary features, this updated understanding gives researchers a starting point to study the relationships between elements and their effective integration into organizational decision processes.

Second, we analyzed the meta-level of PAS, revealing the differing roles of prescriptive agents and human decision-makers from a decision-theoretic perspective and highlighting their synergy, delegation, and adaptability. Our conceptualization led us to derive four archetypes, each providing varying levels of support in the decision-making process. From a practical perspective, our findings can serve as initial blueprints or guiding principles on what system modules and features to consider when designing a PAS for specific purposes.

Third, we identified various technology affordance effects of PAS across various sectors, unveiling the purpose and benefits of employing such systems. Owing to the extensive chronology of leveraging optimization algorithms within industries like manufacturing and transportation, a data-driven approach to prescriptive analytics has experienced considerable exploration in these sectors. Meanwhile, although specific fields exhibit a reduced degree of investigation, a comprehensive assessment revealed a heterogeneous array of research endeavors spanning multiple industries. Finally, our paper reveals six key findings or observations, enabling us to derive various research directions and implications for the BA and IS community.

As with most research endeavors, our study comes with certain limitations. Our goal was to establish the status quo and create a shared understanding of the existing body of knowledge on PAS. To achieve this, we relied on concept matrices as

a qualitative analysis tool, which is never complete and serves as an initial foundation for more comprehensive research and contextualization. Furthermore, we focus our SLR on prescriptive analytics, resulting in an underrepresentation of the socio-technical lens in our literature sample due to the current research's technological and algorithmic focus. However, even though the papers did not explicitly mention the social subsystem or interactions, we could make educated assumptions based on, for example, architectural overviews presented, which enabled us to derive system archetypes, such as how different components synergize with the decision-maker.

Additionally, it is imperative to highlight that our primary objective was to demystify the PAS landscape in research from an overarching IS perspective. We endeavored to conceptualize and categorize the aspects within this domain, acknowledging the inherent challenges in attributing specific concepts to distinct groups, especially for prescriptive analytics techniques. While we recognize that the clarity of separation between these elements is not always absolute, owing to the multifaceted nature and numerous influencing factors, we believe our work provides a substantial foundation. This framework is particularly beneficial for newcomers in the field, whether designing systems or conducting research. It offers a solid grounding, enabling a deeper and more nuanced understanding of the PAS space. We assert that this is essential for anyone aspiring to navigate, contribute to, or innovate within this ever-evolving and complex field. However, additional relevant PAS perspectives may still be explored, for example, within a deeper algorithmic review scope. For instance, a potential area of investigation could be the differentiation between sequential (predict-then-optimize) and simultaneous (predict-and-optimize) PAS. Here, the latter could potentially lead to improved decision outcomes, as it proposes learning a predictive model by directly minimizing the cost of the downstream decision-making task (Vanderschueren et al. 2022; Zhang et al. 2022). Taking this as an example, future reviews with an algorithmic focus would be valuable avenues of inquiry, especially given the recent AI model innovations, which may significantly impact how future PAS are designed.

Further, we note that "prescription" or "prescriptive analytics" may not be used as frequently in every research discipline. Some contributions may only refer to an affordance (e.g., scheduling, routing) or the technique (e.g., optimization, linear programming) in their title, keyword, or abstract, possibly excluding these with our search string. Although prescriptive analytics originated in the BA domain (Holsapple et al. 2014; Delen and Zolbanin 2018), our review revealed that it is a well-established concept in various research disciplines today, with special issues dedicated to the topic in widely respected journals (e.g., Giesecke et al. 2022). It is clear that today, prescriptive analytics is an overarching concept, describing a task objective or even a decision-making paradigm nested in a socio-technical system instead of a specific technique or algorithm to be employed, such as linear programming or a specific ML technique. This trend is also reflected in our sample's growing number of publications after 2018, enabling us to establish a representative view of the current research state with many contemporary examples from the literature. However, our literature-centric approach must be assessed for practical relevance by incorporating real-world PAS applications and engaging with practitioners with hands-on

experience in the field. This perspective will ensure a well-rounded understanding of the topic and its implications and help bridge the gap between theory and practice.

In summary, our research provides a comprehensive understanding of the current state of PAS and highlights areas for future research and development. By exploring these opportunities, researchers and practitioners can collaborate to create more effective and efficient PAS, ultimately driving better decision-making and business value in organizations.

Appendix

Appendix A: Detailed search syntax

Database	Search string
Web of science	TS = ((prescriptive) AND (model OR machine learning OR optimization OR evolutionary OR expert system OR heuristics OR simulation OR artificial intelligence OR analytics))
Scopus	TITLE-ABS-KEY (prescriptive AND (model OR machine AND learning OR optimization OR evolutionary OR expert AND systems OR heuristics OR simulation OR artificial AND intelligence OR analytics))
AISeL	abstract:(prescriptive AND (model OR machine learning OR optimization OR evolutionary OR expert system OR heuristics OR simulation OR artificial intelligence OR analytics)) OR title:(prescriptive AND (model OR machine learning OR optimization OR evolutionary OR expert system OR heuristics OR simulation OR artificial intelligence OR analytics))
ACM DL	[Keywords: prescriptive] AND [[Keywords: analytics] OR [Keywords: model] OR [Keywords: machine learning] OR [Keywords: optimization] OR [Keywords: evolutionary] OR [Keywords: expert system] OR [Keywords: heuristics] OR [Keywords: simulation] OR [Keywords: artificial intelligence]][Title: prescriptive] [[Title: analytics] OR [Title: model] OR [Title: machine learning] OR [Title: optimization] OR [Title: evolutionary] OR [Title: expert system] OR [Title: heuristics] OR [Title: simulation] OR [Title: artificial intelligence]]
IEEE explore	((prescriptive) AND (model OR machine learning OR optimization OR evolutionary OR expert system OR heuristics OR simulation OR artificial intelligence OR analytics))

Appendix B: Literature sample

ID	Year	Article title
1	2020	A data analytic framework for physical fatigue management using wearable sensors
2	2022	A deficiency of prescriptive analytics-No perfect predicted value or predicted distribution exists

ID	Year	Article title
3	2022	A dynamic predict, then optimize preventive maintenance approach using operational intervention data
4	2019	A Dynamic Prescriptive Maintenance Model Considering System Aging and Degradation
5	2020	A Formative Usability Study to Improve Prescriptive Systems for Bioinformatics Big Data
6	2021	A Framework for Pandemic Prediction Using Big Data Analytics
7	2022	A Fuzzy Prescriptive Analytics Approach to Power Generation Capacity Planning
8	2016	A hybrid genetic algorithm-queuing multi-compartment model for optimizing inpatient bed occupancy and associated costs
9	2023	A lower approximation based integrated decision analysis framework for a blockchain-based supply chain
10	2021	A machine learning approach to enable bulk orders of critical spare-parts in the shipping industry
11	2021	A model-based reinforcement learning approach for maintenance optimization of degrading systems in a large state space
12	2020	A Modular Edge-/Cloud-Solution for Automated Error Detection of Industrial Hairpin Weldings using Convolutional Neural Networks
13	2022	A prescriptive analytics approach to employee selection
14	2021	A prescriptive analytics framework for efficient E-commerce order delivery
15	2021	A Prescriptive Analytics Method for Cost Reduction in Clinical Decision Making
16	2022	A prescriptive Dirichlet power allocation policy with deep reinforcement learning
17	2022	A prescriptive framework to support express delivery supply chain expansions in highly urbanized environments
18	2021	A Prescriptive Intelligent System for an Industrial Wastewater Treatment Process: Analyzing pH as a First Approach
19	2022	A Prescriptive Machine Learning Method for Courier Scheduling on Crowdsourced Delivery Platforms
20	2017	A Proactive Event-driven Decision Model for Joint Equipment Predictive Maintenance and Spare Parts Inventory Optimization
21	2017	A procedural approach for realizing prescriptive maintenance planning in manufacturing industries
22	2019	A prognostic algorithm to prescribe improvement measures on throughput bottlenecks
23	2019	A reference architecture based on edge and cloud computing for smart manufacturing
24	2013	A reinforcement learning approach to autonomous decision-making in smart electricity markets
25	2022	A Review of Predictive and Prescriptive Offshore Wind Farm Operation and Maintenance
26	2013	A specialty steel bar company uses analytics to determine available-to-promise dates
27	2021	A Two-Stage Data-Driven Spatiotemporal Analysis to Predict Failure Risk of Urban Sewer Systems Leveraging Machine Learning Algorithms
28	2019	A prescriptive analytics approach to markdown pricing for an e-commerce retailer
29	2021	A prescriptive analytics framework for optimal policy deployment using heterogeneous treatment effects
30	2020	A survey on various applications of prescriptive analytics
31	2013	Adaptive middleware for real-time prescriptive analytics in large scale power systems
32	2019	An artificial intelligence decision support system for unconventional field development design
33	2018	An asset-management oriented methodology for mine haul-fleet usage scheduling

ID	Year	Article title
34	2021	An Automated Tool to Support an Intelligence Learner Management System Using Learning Analytics and Machine Learning
35	2022	An Industry 4.0 Intelligent Decision Support System for Analytical Laboratories
36	2015	An Information System for Sales Team Assignments Utilizing Predictive and Prescriptive Analytics
37	2018	An Integration of Requirement Forecasting and Customer Segmentation Models towards Prescriptive Analytics For Electrical Devices Production
38	2020	An Intelligence Learner Management System using Learning Analytics and Machine learning
39	2022	An Inverse Optimization Approach to Measuring Clinical Pathway Concordance
40	2017	Analysis and optimization based on reusable knowledge base of process performance models
41	2015	Analysis and optimization in smart manufacturing based on a reusable knowledge base for process performance models
42	2023	Analytical Problem Solving Based on Causal, Correlational and Deductive Models
43	2022	Analytics with stochastic optimisation: experimental results of demand uncertainty in process industries
44	2017	Application of derivatives to nonlinear programming for prescriptive analytics
45	2016	Asset health management using predictive and prescriptive analytics for the electric power grid
46	2023	Believing in Analytics: Managers' Adherence to Price Recommendations from a DSS
47	2018	Big data on the shop-floor: sensor-based decision-support for manual processes
48	2021	Bootstrap robust prescriptive analytics
49	2010	Building Business Intelligence Applications Having Prescriptive and Predictive Capabilities
50	2021	Catalyzing a Culture of Care and Innovation Through Prescriptive Analytics and Impact Prediction to Create Full-Cycle Learning
51	2021	Catch me if you scan: Data-driven prescriptive modeling for smart store environments
52	2014	Characterised and personalised predictive-prescriptive analytics using agent-based simulation
53	2019	Chassis Leasing and Selection Policy for Port Operations
54	2020	Closing the loop: Real-time Error Detection and Correction in automotive production using Edge-/Cloud-Architecture and a CNN
55	2021	Condition-based critical level policy for spare parts inventory management
56	2018	Constituent Elements for Prescriptive Analytics Systems
57	2019	Context-aware based restaurant recommender system: A prescriptive analytics
58	2022	Coupled Learning Enabled Stochastic Programming with Endogenous Uncertainty
59	2021	Course performance prediction and evolutionary optimization for undergraduate engineering program towards admission strategic planning
60	2020	Cyber-Physical-Social System for Parallel Driving: From Concept to Application
61	2021	Data Analytics based Prescriptive Analytics for Selection of Lean Manufacturing System
62	2014	Data analytics using simulation for smart manufacturing
63	2018	Data Analytics: The next dimension in molding sand control
64	2011	Data is Dead… Without What-If Models
65	2021	Data-Driven Collaborative Human-AI Decision Making
66	2019	Data-Driven Design Optimization for Industrial Products

ID	Year	Article title
67	2021	Data-Driven Methodology to Support Long-Lasting Logistics and Decision Making for Urban Last-Mile Operations
68	2020	Data-Driven Prescriptive Maintenance: Failure Prediction Using Ensemble Support Vector Classification for Optimal Process and Maintenance Scheduling
69	2021	Decision Support for Knowledge Intensive Processes Using RL Based Recommendations
70	2023	Defining content marketing and its influence on online user behavior: a data-driven prescriptive analytics method
71	2022	Design and Development of We-CDSS Using Django Framework: Conducing Predictive and Prescriptive Analytics for Coronary Artery Disease
72	2020	Design and Evaluation of a Process-aware Recommender System based on Prescriptive Analytics
73	2015	Design and Implementation of the LogicBlox System
74	2022	Developing a prescriptive decision support system for shop floor control
75	2018	Differentially Private Prescriptive Analytics
76	2022	Dynamic Pricing for New Products Using a Utility-Based Generalization of the Bass Diffusion Model
77	2020	Dynamic Thresholding Leading to Optimal Inventory Maintenance
78	2016	Early Predictions of Movie Success: The Who, What, and When of Profitability
79	2020	Effective reinforcement learning through evolutionary surrogate-assisted prescription
80	2020	Employees recruitment: A prescriptive analytics approach via machine learning and mathematical programming
81	2019	Evaluation of the Selected Philippine E-Government Websites' Performance with Prescriptive Analysis
82	2016	EventAction: Visual analytics for temporal event sequence recommendation
83	2020	Expert-in-the-loop prescriptive analytics using mobility intervention for epidemics
84	2022	Explainable Process Prescriptive Analytics
85	2017	Fast integrated reservoir modelling on the Gjøa field offshore Norway
86	2019	Fault Classification and Correction based on Convolutional Neural Networks exemplified by laser welding of hairpin windings
87	2013	Five pillars of prescriptive analytics success
88	2016	Fleet asset capacity analysis and revenue management optimization using advanced prescriptive analytics
89	2016	Fossil-free public transport: Prescriptive policy analysis for the Swedish bus fleets
90	2018	France's Governmental Big Data Analytics: From Predictive to Prescriptive Using R
91	2023	From "prepare for the unknown" to "train for what's coming": A digital twin-driven and cognitive training approach for the workforce of the future in smart factories
92	2021	From Prediction to Prescription: Evolutionary Optimization of Nonpharmaceutical Interventions in the COVID-19 Pandemic
93	2020	From predictive to prescriptive analytics
94	2020	From predictive to prescriptive analytics: A data-driven multi-item newsvendor model
95	2020	From predictive to prescriptive process monitoring: Recommending the next best actions instead of calculating the next most likely events
96	2015	From predictive uplift modeling to prescriptive uplift analytics: A practical approach to treatment optimization while accounting for estimation risk
97	2023	Fundamental challenge and solution methods in prescriptive analytics for freight transportation

ID	Year	Article title
98	2020	HadoopSec 2.0: Prescriptive analytics-based multi-model sensitivity-aware constraints centric block placement strategy for Hadoop
99	2020	How prescriptive analytics influences decision making in precision medicine
100	2021	Human-augmented prescriptive analytics with interactive multi-objective reinforcement learning
101	2020	Hybrid Data-Driven and Physics-Based Modeling for Gas Turbine Prescriptive Analytics
102	2022	Hybrid Neuro-Genetic Machine Learning Models for the Engineering of Ring-spun Cotton Yarns
103	2017	Identifying cost-effective waterflooding optimization opportunities in mature reservoirs from data driven analytics
104	2017	Impact of Business Analytics and Enterprise Systems on Managerial Accounting
105	2020	Improving harvesting operations in an oil palm plantation
106	2022	Improving Prescriptive Maintenance by Incorporating Post-Prognostic Information Through Chance Constraints
107	2022	Improving the tactical planning of solid waste collection with prescriptive analytics: a case study
108	2022	Improving Variable Orderings of Approximate Decision Diagrams Using Reinforcement Learning
109	2019	Innovative InterLabs System for Smart Learning Analytics in Engineering Education
110	2018	Integrated assortment planning and store-wide shelf space allocation: An optimization-based approach
111	2018	Integrative Analytics for Detecting and Disrupting Transnational Interdependent Criminal Smuggling, Money, and Money-Laundering Networks
112	2011	Intelligent call routing: Optimizing contact center throughput
113	2021	Intervention Support Program for Students at Risk of Dropping Out Using Fuzzy Logic-Based Prescriptive Analytics
114	2023	Inventory Waste Management with Augmented Analytics for Finished Goods
115	2021	JANOS: An Integrated Predictive and Prescriptive Modeling Framework
116	2014	Key Performance Indicators for Facility Performance Assessment: Simulation of Core Indicators
117	2022	Landscape Optimization for Prescribed Burns in Wildfire Mitigation Planning
118	2020	Layered Behavior Modeling via Combining Descriptive and Prescriptive Approaches: A Case Study of Infantry Company Engagement
119	2019	Learning failure modes of soil slopes using monitoring data
120	2019	Learning to Match via Inverse Optimal Transport
121	2022	Linking Predictive and Prescriptive Analytics of Elderly and Frail Patient Hospital Services
122	2020	Location-based social simulation for prescriptive analytics of disease spread
123	2020	Machine Learning for Predictive and Prescriptive Analytics of Operational Data in Smart Manufacturing
124	2021	Managing the Training Process in Elite Sports: From Descriptive to Prescriptive Data Analytics
125	2015	Marketing Strategy Support System for Small Businesses
126	2015	Media company uses analytics to schedule radio advertisement spots
127	2013	Model-based decision support for optimal brochure pricing: applying advanced analytics in the tour operating industry
128	2020	Model-predictive safety optimal actions to detect and handle process operation hazards
129	2022	Network Analytics for Infrastructure Asset Management Systemic Risk Assessment

ID	Year	Article title
130	2015	Nurse-patient assignment models considering patient acuity metrics and nurses' perceived workload
131	2019	Offline-Online Approximate Dynamic Programming for Dynamic Vehicle Routing with Stochastic Requests
132	2017	On the adoption and impact of predictive analytics for server incident reduction
133	2022	Operations (management) warp speed: Rapid deployment of hospital-focused predictive/prescriptive analytics for the COVID-19 pandemic
134	2022	Optimal policy trees
135	2017	Optimization Beyond Prediction: Prescriptive Price Optimization
136	2018	Optimized assignment patterns in Mobile Edge Cloud networks
137	2020	Optimized Maintenance Decision-Making—A Simulation-Supported Prescriptive Analytics Approach Based on Probabilistic Cost–Benefit Analysis
138	2022	Optimizing diesel fuel supply chain operations to mitigate power outages for hurricane relief
139	2018	Optimizing outpatient appointment system using machine learning algorithms and scheduling rules: A prescriptive analytics framework
140	2023	Optimizing the preventive maintenance frequency with causal machine
141	2015	People Skills: Building Analytics Decision Models That Managers Use-A Change Management Perspective
142	2021	Personalized Multimorbidity Management for Patients with Type 2 Diabetes Using Reinforcement Learning of Electronic Health Records
143	2017	Petroleum Analytics Learning Machine' for optimizing the Internet of Things of today's digital oil field-to-refinery petroleum system
144	2021	PI prob: A risk prediction and clinical guidance system for evaluating patients with recurrent infections
145	2022	Pitfalls and protocols of data science in manufacturing practice
146	2022	Predict, then schedule: Prescriptive analytics approach for machine learning-enabled sequential clinical scheduling
147	2022	Predicting biodiesel properties and its optimal fatty acid profile via explainable machine learning
148	2016	Predictive analytics model for healthcare planning and scheduling
149	2019	Predictive and Prescriptive Analytics for Performance Optimization: Framework and a Case Study on a Large-Scale Enterprise System
150	2021	Predictive and prescriptive analytics in transportation geotechnics: Three case studies
151	2023	Predictive and Prescriptive Business Process Monitoring with Reinforcement Learning
152	2019	Predictive and prescriptive analytics for location selection of add-on retail products
153	2017	Predictive and prescriptive analytics, machine learning and child welfare risk assessment: The Broward County experience
154	2022	Predictive machine learning for prescriptive applications: A coupled training–validating approach
155	2019	Predictive, prescriptive and detective analytics for smart manufacturing in the information age
156	2020	Prescriptive Analytics Aids Completion Optimization in Unconventionals
157	2023	Prescriptive analytics applications in sustainable operations research: conceptual framework and future research challenges
158	2015	Prescriptive Analytics Applied to Brace Treatment for AIS: A Pilot Demonstration

ID	Year	Article title
159	2015	Prescriptive Analytics Based Autonomic Networking for Urban Streams Services Provisioning
160	2023	Prescriptive analytics for a multi-shift staffing problem
161	2013	Prescriptive Analytics for Allocating Sales Teams to Opportunities
162	2016	Prescriptive analytics for big data
163	2019	Prescriptive analytics for completion optimization in unconventional resources
164	2017	Prescriptive analytics for FIFA World Cup lodging capacity planning
165	2022	Prescriptive Analytics for finding the optimal manufacturing practice based on the simulation models of Lean Manufacturing and Total Quality Management
166	2022	Prescriptive Analytics for Flexible Capacity Management
167	2019	Prescriptive analytics for human resource planning in the professional services industry
168	2021	Prescriptive analytics for impulsive behaviour prevention using real-time biometrics
169	2018	Prescriptive Analytics for MEC Orchestration
170	2015	Prescriptive analytics for planning research-performance strategy
171	2014	Prescriptive analytics for recommendation-based business process optimization
172	2020	Prescriptive analytics for reducing 30-day hospital readmissions after general surgery
173	2020	Prescriptive Analytics for Swapping Aircraft Assignments at All Nippon Airways
174	2016	Prescriptive analytics for understanding of out-of-plane deformation in additive manufacturing
175	2018	Prescriptive analytics in airline operations: arrival time prediction and cost index optimization for short-haul flights
176	2021	Prescriptive Analytics in Internet of Things with Concentration on Deep Learning
177	2022	Prescriptive Analytics in Procurement: Reducing Process Costs
178	2021	Prescriptive analytics in public-sector decision-making: A framework and insights from charging infrastructure planning
179	2013	Prescriptive Analytics System for Improving Research Power
180	2014	Prescriptive analytics system for scholar research performance enhancement
181	2018	Prescriptive analytics through constrained Bayesian optimization
182	2021	Prescriptive analytics with differential privacy
183	2021	Prescriptive Analytics for Dynamic Real Time Scheduling of Diffusion Furnaces
184	2021	Prescriptive analytics for flexible capacity management
185	2020	Prescriptive analytics for inventory management in health care
186	2020	Prescriptive Analytics for Real-Time Optimization of Deepwater Casing Exits
187	2021	Prescriptive Analytics in Urban Policing Operations
188	2015	Prescriptive analytics using synthetic information
189	2019	Prescriptive analytics: a survey of emerging trends and technologies
190	2020	Prescriptive analytics: Literature review and research challenges
191	2022	Prescriptive block replacement policy for production degrading systems
192	2020	Prescriptive business process monitoring for recommending next best actions
193	2019	Prescriptive cluster-dependent support vector machines with an application to reducing hospital readmissions
194	2016	Prescriptive Control of Business Processes
195	2020	Prescriptive data analytics to optimize casing exits
196	2019	Prescriptive Equipment Maintenance: A Framework
197	2022	Prescriptive Healthcare Analytics: A Tutorial on Discrete Optimization and Simulation

ID	Year	Article title
198	2014	Prescriptive information fusion
199	2020	Prescriptive Learning for Air-Cargo Revenue Management
200	2019	Prescriptive Maintenance of Railway Infrastructure: From Data Analytics to Decision Support
201	2022	Prescriptive maintenance technique for photovoltaic systems
202	2020	Prescriptive Modelling System Design for an Armature Multi-coil Rewinding Cobot Machine
203	2020	Prescriptive Process Analytics with Deep Learning and Explainable Artificial Intelligence
204	2022	Prescriptive process monitoring: Quo vadis?
205	2023	Prescriptive selection of machine learning hyperparameters with applications in power markets: Retailer's optimal trading
206	2022	Prescriptive Trees for Integrated Forecasting and Optimization Applied in Trading of Renewable Energy
207	2018	Prescriptive analytics system for long-range aircraft conflict detection and resolution
208	2017	Prescstream: A framework for streaming soft real-time predictive and prescriptive analytics
209	2020	Price Investment using Prescriptive Analytics and Optimization in Retail
210	2019	PriMa: a prescriptive maintenance model for cyber-physical production systems
211	2022	PROAD (Process Advisor): A health monitoring framework for centrifugal pumps
212	2021	Probation Status Prediction and Optimization for Undergraduate Engineering Students
213	2018	Product Portfolio Design Using Prescriptive Analytics
214	2019	Rapid spatio-temporal flood prediction and uncertainty quantification using a deep learning method
215	2016	Realtime Predictive and Prescriptive Analytics with Real-time Data and Simulation
216	2022	Reducing the Total Product Cost at the Product Design Stage
217	2020	Replenishment and denomination mix of automated teller machines with dynamic forecast demands
218	2020	Requirements for Prescriptive Recommender Systems Extending the Lifetime of EV Batteries
219	2014	Research Advising System Based on Prescriptive Analytics
220	2018	Rh-rt: A Data Analytics Framework for Reducing Wait Time at Emergency Departments and Centres for Urgent Care
221	2023	Rollout-based routing strategies with embedded prediction: A fish trawling application
222	2019	Route-cost-assignment with joint user and operator behavior as a many-to-one stable matching assignment game
223	2018	Rules engine and complex event processor in the context of internet of things for precision agriculture
224	2019	SAFE: A Comprehensive Data Visualization System
225	2021	Seat Assignments With Physical Distancing in Single-Destination Public Transit Settings
226	2022	Selecting advanced analytics in manufacturing: a decision support model
227	2020	Sensor-Driven Learning of Time-Dependent Parameters for Prescriptive Analytics
228	2015	Service-Delivery Modeling and Optimization
229	2020	Simulation as a decision-making tool in a business analytics environment
230	2021	Smart "Predict, then Optimize"
231	2017	Smart Maintenance Decision Support Systems (SMDSS) based on corporate big data analytics
232	2019	Smart Manufacturing with Prescriptive Analytics

ID	Year	Article title
233	2022	Smart urban transport and logistics: A business analytics perspective
234	2016	Social media optimization: Identifying an optimal strategy for increasing network size on Facebook
235	2017	Software estimation—towards prescriptive analytics
236	2021	SolveDB + : SQL-based prescriptive analytics
237	2022	Solving an Instance of a Routing Problem Through Reinforcement Learning and High Performance Computing
238	2014	Sonora: A Prescriptive Model for Message Authoring on Twitter
239	2022	Spare parts supply with incoming quality control and inspection errors in condition based maintenance
240	2022	Stochastic dynamic vehicle routing in the light of prescriptive analytics: A review
241	2021	Stock market predictor using prescriptive analytics
242	2014	System Thinking: Crafting Scenarios for Prescriptive Analytics
243	2023	The Analytics of Bed Shortages: Coherent Metric, Prediction, and Optimization
244	2017	The green fleet optimization model for a low-carbon economy: A prescriptive analytics
245	2023	The Impact of Dashboard Feedback Type on Learning Effectiveness, Focusing on Learner Differences
246	2021	The Methodology of Hybrid Modelling for Gas Turbine Subsystems Prescriptive Analytics
247	2022	The role of optimization in some recent advances in data-driven decision-making
248	2021	To imprison or not to imprison: an analytics model for drug courts
249	2019	Topical Prescriptive Analytics System for Automatic Recommendation of Convergence Technology
250	2019	Towards an automated optimization-as-a-service concept
251	2022	Uncertainty-bounded reinforcement learning for revenue optimization in air cargo: a prescriptive learning approach
252	2023	University admission process: a prescriptive analytics approach
253	2019	Unleashing Analytics to Reduce Costs and Improve Quality in Wastewater Treatment
254	2018	Using Bayesian belief network and time-series model to conduct prescriptive and predictive analytics for computer industries
255	2019	Using Prescriptive Data Analytics to Reduce Grading Bias and Foster Student Success
256	2020	Using prescriptive analytics for the determination of optimal crop yield
257	2019	Using prescriptive analytics to support the continuous improvement process
258	2020	Verizon Uses Advanced Analytics to Rationalize Its Tail Spend Suppliers
259	2022	Virtual Material Quality Investigation System
260	2019	Visual PROMETHEE: Developments of the PROMETHEE & GAIA multicriteria decision aid methods
261	2021	Wartime industrial logistics information integration: Framework and application in optimizing deployment and formation of military logistics platforms
262	2019	What to do when decision-makers deviate from model recommendations? Empirical evidence from hydropower industry

Appendix C: Identified constituent components in literature sample

Constituent components	References from the literature sample discussing or mentioning the concepts	Quantity
Decision variables	[2], [3], [4], [7], [8], [9], [10], [11], [13], [14], [15], [17], [18], [19], [20], [25], [26], [27], [28], [29], [30], [33], [36], [39], [40], [42], [43], [44], [45], [48], [53], [56], [58], [59], [65], [66], [67], [68], [69], [73], [74], [75], [76], [79], [80], [83], [84], [87], [90], [91], [92], [93], [94], [96], [97], [100], [102], [103], [105], [106], [107], [108], [110], [112], [115], [117], [120], [123], [126], [127], [128], [129], [130], [131], [135], [136], [137], [138], [139], [140], [145], [146], [147], [150], [152], [154], [156], [160], [161], [163], [164], [165], [166], [167], [169], [170], [173], [175], [176], [177], [178], [181], [182], [183], [184], [185], [186], [187], [189], [190], [191], [193], [194], [195], [198], [199], [200], [201], [202], [205], [206], [207], [209], [212], [216], [217], [221], [222], [225], [226], [230], [234], [236], [237], [239], [240], [242], [243], [244], [247], [250], [252], [253], [258], [259], [261], [262]	147
Objectives	[2], [3], [4], [7], [8], [9], [10], [11], [13], [14], [15], [17], [18], [19], [20], [24], [25], [26], [27], [28], [29], [30], [33], [36], [39], [40], [42], [43], [44], [45], [48], [49], [53], [56], [58], [59], [60], [65], [66], [67], [68], [69], [73], [74], [75], [76], [79], [80], [83], [84], [87], [90], [91], [92], [93], [94], [96], [97], [100], [102], [103], [104], [105], [106], [107], [108], [110], [112], [115], [117], [120], [123], [126], [127], [128], [129], [130], [131], [135], [136], [137], [138], [139], [140], [143], [145], [146], [147], [149], [150], [151], [152], [154], [155], [156], [160], [161], [162], [163], [164], [166], [167], [169], [170], [173], [175], [176], [177], [178], [181], [182], [183], [184], [185], [186], [187], [189], [190], [191], [193], [194], [195], [198], [199], [200], [201], [202], [204], [205], [206], [207], [209], [212], [213], [216], [217], [221], [222], [225], [226], [227], [229], [230], [234], [236], [237], [239], [240], [242], [243], [244], [247], [250], [252], [258], [259], [261], [262]	158

Constituent components	References from the literature sample discussing or mentioning the concepts	Quantity
Constraints	[2], [3], [4], [7], [8], [13], [14], [15], [17], [19], [25], [26], [27], [28], [29], [30], [33], [36], [39], [40], [42], [43], [44], [45], [48], [49], [53], [56], [58], [65], [66], [67], [68], [69], [72], [73], [74], [75], [76], [80], [84], [87], [90], [91], [92], [93], [94], [96], [97], [100], [102], [103], [104], [105], [106], [107], [108], [110], [112], [115], [117], [120], [123], [126], [127], [128], [129], [130], [135], [136], [137], [138], [139], [140], [145], [146], [147], [152], [154], [155], [156], [160], [161], [162], [163], [164], [166], [167], [169], [170], [173], [175], [176], [177], [178], [181], [182], [183], [184], [185], [186], [187], [189], [190], [191], [195], [198], [199], [200], [201], [202], [205], [206], [207], [209], [216], [217], [221], [222], [225], [226], [227], [230], [234], [236], [239], [240], [242], [243], [244], [247], [250], [253], [258], [259], [261], [262]	137
Current state	[1], [3], [4], [6], [9], [10], [11], [15], [16], [17], [20], [23], [24], [25], [26], [27], [30], [34], [35], [36], [37], [38], [40], [41], [42], [45], [50], [52], [53], [56], [59], [60], [65], [66], [67], [68], [69], [70], [78], [80], [82], [90], [93], [94], [96], [97], [99], [100], [103], [104], [105], [107], [108], [109], [110], [111], [112], [113], [114], [117], [118], [123], [125], [127], [129], [130], [132], [133], [136], [137], [138], [139], [140], [145], [146], [147], [152], [155], [162], [163], [166], [169], [173], [176], [178], [179], [180], [182], [184], [185], [186], [189], [190], [191], [192], [194], [195], [196], [200], [202], [205], [207], [209], [210], [211], [212], [216], [217], [218], [219], [220], [221], [223], [224], [226], [228], [229], [231], [233], [235], [236], [240], [241], [243], [247], [248], [249], [253], [257], [258], [259], [261], [262]	133

Constituent components	References from the literature sample discussing or mentioning the concepts	Quantity
Probabilities	[1], [2], [3], [6], [7], [9], [10], [11], [12], [13], [14], [15], [17], [18], [19], [20], [22], [24], [25], [27], [28], [29], [30], [31], [32], [34], [35], [36], [37], [38], [39], [40], [41], [42], [45], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [63], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [77], [78], [79], [80], [82], [84], [86], [87], [88], [90], [91], [92], [93], [94], [95], [96], [98], [99], [100], [101], [102], [103], [104], [105], [106], [108], [109], [110], [111], [112], [114], [115], [119], [120], [121], [123], [125], [127], [128], [129], [131], [132], [133], [134], [135], [137], [138], [139], [140], [143], [144], [145], [146], [147], [148], [149], [151], [152], [153], [154], [155], [156], [158], [160], [161], [162], [163], [164], [166], [167], [168], [171], [172], [175], [176], [177], [178], [181], [182], [184], [185], [186], [187], [189], [190], [191], [192], [193], [194], [195], [196], [198], [199], [202], [203], [204], [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [218], [220], [221], [223], [224], [226], [228], [229], [230], [231], [233], [234], [235], [236], [238], [239], [240], [241], [246], [247], [248], [251], [252], [253], [254], [256], [257], [258], [259], [261], [262]	201
Mathematical program	[2], [3], [4], [7], [9], [10], [13], [14], [15], [16], [17], [19], [20], [25], [27], [28], [29], [30], [32], [33], [35], [36], [39], [40], [41], [42], [43], [44], [45], [47], [48], [49], [51], [53], [56], [58], [65], [67], [68], [73], [74], [75], [76], [80], [81], [83], [84], [87], [88], [89], [93], [94], [96], [97], [99], [100], [101], [103], [104], [105], [106], [107], [110], [112], [115], [117], [120], [121], [124], [126], [127], [129], [130], [131], [135], [136], [138], [140], [146], [149], [150], [152], [154], [155], [156], [157], [160], [161], [162], [163], [164], [166], [167], [169], [170], [173], [174], [175], [176], [177], [178], [181], [182], [183], [184], [185], [186], [187], [189], [190], [191], [194], [195], [197], [198], [199], [200], [201], [204], [205], [206], [207], [209], [210], [216], [217], [218], [220], [221], [222], [225], [226], [228], [229], [230], [233], [236], [239], [240], [243], [247], [250], [251], [252], [253], [258], [260], [261], [262]	149

Constituent components	References from the literature sample discussing or mentioning the concepts	Quantity
Machine learning	[1], [2], [6], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [23], [24], [25], [26], [27], [29], [30], [32], [34], [35], [36], [37], [40], [41], [42], [45], [48], [50], [51], [53], [54], [56], [57], [58], [59], [60], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [77], [78], [79], [80], [83], [84], [86], [88], [90], [92], [93], [94], [95], [96], [98], [99], [100], [101], [102], [103], [104], [105], [106], [108], [109], [112], [114], [115], [120], [121], [123], [127], [129], [132], [133], [134], [135], [137], [139], [140], [142], [143], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [159], [160], [161], [162], [163], [166], [168], [169], [171], [172], [175], [176], [182], [184], [185], [186], [187], [189], [190], [192], [193], [195], [196], [198], [199], [202], [203], [204], [205], [206], [208], [209], [210], [211], [212], [213], [214], [216], [218], [220], [221], [224], [226], [229], [230], [233], [234], [236], [237], [238], [240], [241], [246], [247], [248], [250], [251], [252], [253], [254], [255], [256], [258], [259]	171
Evolutionary comp	[8], [18], [25], [37], [59], [66], [67], [83], [90], [92], [102], [104], [117], [128], [145], [147], [156], [176], [181], [190], [212], [226], [229], [233], [234], [236], [259]	27
Simulation	[17], [19], [21], [30], [40], [41], [52], [53], [56], [61], [62], [64], [67], [72], [74], [83], [87], [88], [89], [91], [92], [99], [101], [103], [104], [105], [107], [109], [116], [117], [122], [132], [133], [135], [137], [139], [143], [157], [158], [160], [162], [172], [176], [188], [189], [190], [192], [197], [198], [199], [205], [210], [213], [215], [216], [220], [224], [226], [228], [229], [233], [238], [243], [250], [254], [259], [261]	67
Logic-based models	[90], [113], [118], [165], [171], [190], [204], [211], [226], [229], [231], [233]	12
Probabilistic models	[3], [21], [25], [27], [50], [55], [73], [75], [90], [104], [110], [118], [119], [129], [137], [144], [145], [151], [158], [168], [177], [189], [190], [196], [198], [200], [204], [207], [210], [216], [220], [221], [226], [227], [228], [229], [233], [254]	38

Constituent components	References from the literature sample discussing or mentioning the concepts	Quantity
Single decision	[[1], [2], [3], [4], [7], [8], [9], [10], [11], [13], [14], [15], [16], [17], [19], [20], [26], [27], [28], [29], [30], [36], [37], [39], [40], [42], [43], [45], [49], [53], [54], [56], [58], [65], [66], [67], [68], [69], [74], [76], [79], [80], [84], [86], [89], [91], [92], [93], [94], [96], [98], [100], [102], [103], [105], [106], [107], [108], [110], [112], [114], [115], [117], [118], [119], [123], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138], [140], [142], [146], [147], [148], [149], [154], [160], [161], [163], [164], [165], [167], [168], [169], [171], [173], [174], [175], [177], [178], [181], [183], [185], [187], [190], [192], [193], [194], [195], [199], [200], [202], [203], [205], [206], [207], [209], [211], [216], [217], [221], [222], [225], [226], [227], [228], [230], [234], [237], [239], [240], [243], [244], [247], [251], [252], [253], [258], [259], [262]	128
Multiple decisions	[4], [40], [84], [91], [92], [105], [112], [114], [115], [134], [136], [151], [169], [190], [209], [211], [213], [216], [222], [231], [257]	21
execution	[16], [21], [23], [24], [51], [54], [60], [67], [77], [82], [91], [98], [112], [128], [136], [155], [159], [163], [192], [196], [199], [202], [210], [223], [251]	25
Adaptation	[4], [11], [12], [16], [21], [24], [32], [40], [51], [54], [60], [65], [67], [77], [79], [82], [83], [92], [100], [114], [123], [136], [159], [162], [194], [196], [198], [210], [218], [220], [227], [234], [250], [262]	34
Integration	[9], [12], [23], [26], [30], [31], [34], [35], [36], [38], [40], [41], [49], [54], [59], [60], [64], [65], [67], [74], [77], [100], [112], [114], [123], [125], [129], [132], [143], [145], [159], [162], [163], [173], [189], [190], [194], [196], [208], [210], [211], [223], [227], [228], [253], [257], [258], [261]	48
Distributed computing	[12], [23], [30], [31], [54], [114], [162], [189], [190], [194], [196], [208], [209], [210], [223], [236]	16
Modulization	[6], [12], [14], [21], [23], [25], [26], [30], [32], [34], [35], [36], [38], [40], [41], [45], [47], [54], [56], [67], [68], [71], [72], [74], [77], [79], [80], [91], [92], [98], [104], [105], [109], [111], [112], [113], [114], [117], [118], [123], [125], [128], [129], [132], [133], [136], [139], [145], [146], [147], [149], [162], [163], [189], [190], [194], [196], [208], [209], [210], [211], [214], [216], [219], [220], [223], [224], [227], [228], [229], [231], [233], [234], [249], [251], [252], [253], [258], [259], [261]	80
Security- and privacy-preserving	[34], [38], [59], [75], [82], [98], [159], [182], [190], [210]	10
Workflow interface	[5], [6], [9], [12], [30], [32], [34], [36], [38], [40], [41], [49], [65], [67], [73], [82], [91], [92], [100], [104], [114], [115], [123], [163], [173], [189], [190], [194], [210], [216], [218], [219], [220], [223], [225], [227], [228], [236], [250], [251], [257], [258], [262]	43
Explainability	[84], [147], [203]	3

Constituent components	References from the literature sample discussing or mentioning the concepts	Quantity
Visualization	[3], [5], [6], [9], [12], [21], [23], [34], [35], [36], [38], [40], [41], [45], [46], [49], [56], [64], [65], [67], [70], [71], [77], [79], [82], [84], [90], [91], [100], [103], [104], [109], [114], [115], [117], [123], [129], [132], [143], [145], [162], [163], [173], [179], [180], [189], [190], [194], [195], [196], [200], [203], [207], [209], [210], [211], [216], [219], [221], [224], [225], [228], [229], [236], [241], [245], [256], [257], [258], [260], [261]	71
Extensibility	[30], [40], [49], [73], [189], [208], [218], [236]	8

Appendix D: Identified technology affordances in literature sample

Affordance (effect); frequency	Example studies (IDs)
Maintenance planning for optimal maintenance schedule; n = 24	[3], [4], [11], [12], [20], [21], [22], [54], [55], [68], [77], [86], [101], [106], [137], [140], [171], [191], [196], [210], [211], [231], [239], [246]
Production planning for optimal manufacturing schedule; n = 13	[26], [37], [43], [47], [61], [74], [102], [123], [165], [183], [226], [257], [259]
Product (portfolio) design optimization; n = 4	[66], [202], [213], [216]
Operations safety improvement and planning; n = 1	[128]
Industrial worker training optimization; n = 1	[91]
Deformation control in additive manufacturing; n = 1	[174]
Optimization of routing and scheduling; n = 15	[17], [19], [60], [67], [89], [97], [131], [159], [175], [207], [222], [233], [237], [240], [244]
Capacity/cargo management and improvement; n = 8	[53], [88], [166], [173], [184], [199], [225], [251]
Vehicle maintenance planning; n = 2	[10], [200]
Patient treatment planning and improvement; n = 8	[39], [99], [121], [142], [144], [158], [172], [193]
Patient scheduling; n = 8	[8], [130], [139], [146], [148], [197], [220], [243]
Pandemic/epidemic intervention planning; n = 5	[6], [83], [92], [122], [133]
Human health tracking and improvement; n = 4	[1], [71], [168]
Assortment and inventory planning (health); n = 1	[158]
Clinical investment management; n = 1	[15]
Optimizing power system/grid operations; n = 5	[7], [16], [25], [31], [262]
Disaster preparation/recovery planning; n = 3	[117], [138], [214]
Electricity brokerage optimization; n = 3	[24], [205], [206]
Maintenance planning (energy & environment); n = 3	[27], [45], [201]
Wastewater treatment improvement; n = 2	[18], [253]
Waste collection and management planning; n = 2	[107], [114]
Optimization of deepwater casing exits; n = 2	[186], [195]

Affordance (effect); frequency	Example studies (IDs)
Waterflooding process optimization; n = 1	[103]
Battery lifetime optimization; n = 1	[218]
Reservoir design planning; n = 1	[32]
Soil slope analysis; n = 1	[119]
Price optimization; n = 4	[46], [76], [135], [209]
Assortment and inventory planning; n = 3	[94], [110], [152]
Sales team assignments; n = 2	[36], [161]
Customer characterization; n = 1	[52]
Customer service recommendation; n = 1	[96]
Theft surveillance and automated checkout; n = 1	[51]
Academic performance improving; n = 6	[34], [38], [50], [109], [245], [255]
Dropout prevention planning; n = 2	[113], [212]
Admissions planning and selection; n = 2	[59], [252]
Maximize oil/gas recovery; n = 4	[85], [143], [156], [163]
Mining fleet scheduling; n = 1	[33]
Sand molding process improvement; n = 1	[63]
Laboratory task allocation and planning; n = 1	[35]
Biodiesel properties optimization; n = 1	[147]
Social media usage optimization; n = 2	[234], [238]
Network and computing resource orchestration; n = 2	[136], [169]
Software development estimation; n = 1	[235]
Website performance analysis and optimization; n = 1	[81]
Research advising; n = 6	[170], [179], [180], [219], [242], [249]
Harvesting operations planning and optimization, n = 3	[90], [105], [223]
Crop yield optimization; n = 1	[256]
Fish trawling routing and optimization; n = 1	[221]
Law enforcement resource allocation and planning; n = 2	[111], [187]
Imprisonment decision planning and recommendation; n = 1	[164]
Tournament lodging planning; n = 1	[248]
Sports event safety management and planning; n = 1	[224]
Athlete training process improvement; n = 1	[124]
Infantry engagement planning; n = 1	[118]
Military logistics planning; n = 1	[261]
Markdown planning and price optimization; n = 1	[28]
Oder delivery scheduling; n = 1	[14]
Teller machine replenishment planning and allocation; n = 1	[217]
Stock purchase recommendations; n = 1	[241]

Affordance (effect); frequency	Example studies (IDs)
Radio advertising scheduling; n = 1	[126]
Movie planning and profit-maximizing; n = 1	[78]
Project staffing planning and allocation; n = 2	[167], [228]
Child welfare assessment; n = 1	[153]
Infrastructure planning and optimization; n = 1	[178]
Optimal tour pricing; n = 1	[127]
Restaurant recommendations; n = 1	[57]
Prescriptive process management; n = 8	[40], [84], [95], [151], [192], [194], [203], [204]
Employee recruiting and staffing; n = 3	[13], [80], [160]
Procurement and supplier management; n = 2	[177], [258]
Marketing management; n = 2	[70], [125]
Facility/asset management; n = 2	[116], [129]
Managerial Accounting; n = 1	[104]
Call center routing; n = 1	[112]
Server incident management and prevention; n = 1	[132]

Appendix E: Identified system archetypes

System archetypes	References from the literature sample
Advisory PAS	[1], [2], [3], [5], [6], [7], [8], [9], [10], [13], [14], [15], [17], [18], [19], [20], [22], [26], [27], [28], [29], [31], [33], [34], [35], [36], [37], [38], [39], [41], [42], [43], [44], [45], [47], [48], [49], [52], [53], [55], [57], [58], [59], [61], [62], [63], [64], [66], [68], [69], [70], [71], [72], [73], [74], [75], [76], [78], [80], [81], [84], [85], [86], [88], [89], [90], [93], [94], [95], [96], [97], [99], [101], [102], [103], [104], [105], [106], [107], [109], [110], [111], [113], [115], [116], [117], [118], [119], [120], [121], [122], [124], [125], [126], [127], [129], [130], [131], [132], [133], [134], [135], [137], [138], [139], [140], [142], [143], [144], [146], [147], [148], [149], [151], [152], [153], [154], [156], [158], [160], [161], [164], [165], [166], [167], [168], [169], [170], [171], [172], [173], [174], [175], [177], [178], [179], [180], [181], [182], [183], [184], [185], [186], [187], [188], [191], [193], [195], [197], [200], [201], [203], [205], [206], [207], [208], [209], [211], [212], [213], [214], [215], [216], [217], [219], [221], [222], [224], [225], [226], [228], [229], [230], [231], [236], [237], [238], [239], [240], [241], [242], [243], [244], [246], [247], [248], [249], [252], [253], [254], [255], [256], [257], [258], [259], [260], [261]
Executive PAS	[23], [91], [98], [112], [128], [155], [163], [192], [199], [202], [223], [251]
Adaptive PAS	[4], [11], [12], [32], [40], [65], [79], [83], [92], [100], [114], [123], [162], [194], [198], [218], [220], [227], [234], [250], [262]
Self-governing PAS	[16], [21] [24], [51], [54], [60], [67], [77], [82], [136], [159], [196], [210]

Appendix F: Decision processing techniques

In the following, we give an overview of exemplary technique subcategories used in prescriptive analytics, following the survey of Lepenioti et al. (2020).

Decision processing	Example techniques
Mathematical programming	Mixed integer programming, linear programming, binary quadratic programming, non-linear programming, stochastic optimization, conditional stochastic optimization, constrained Bayesian optimization, fuzzy linear programming, dynamic programming
Machine learning	Various clustering algorithms, reinforcement learning, Boltzmann machine, (deep) artificial neural networks
Evolutionary computation	Genetic algorithms, evolutionary optimization, greedy algorithms, particle swarm optimization
Simulation	Simulation over random forest, risk assessments, stochastic simulations, what-if scenarios
Logic-based models	Association rules, decision rules, criteria-based rules, fuzzy rules, distributed rules, benchmark rules, desirability functions, graph-based recommendations
Probabilistic models	Markov decision processes, hidden Markov models, Markov chains

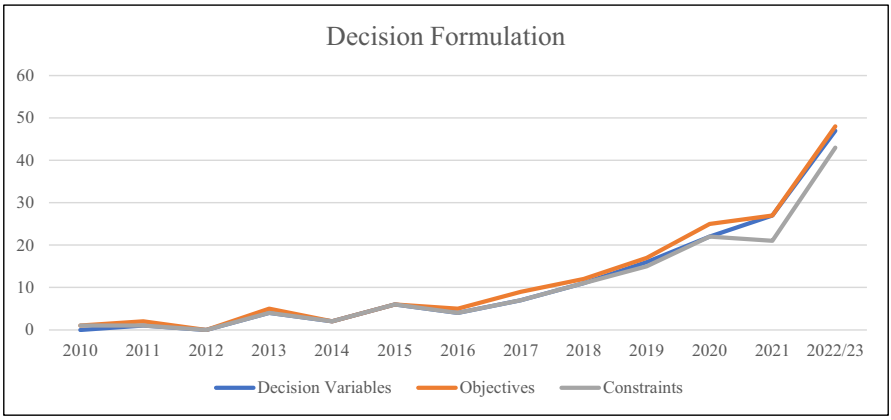
Appendix G: Overview of data properties

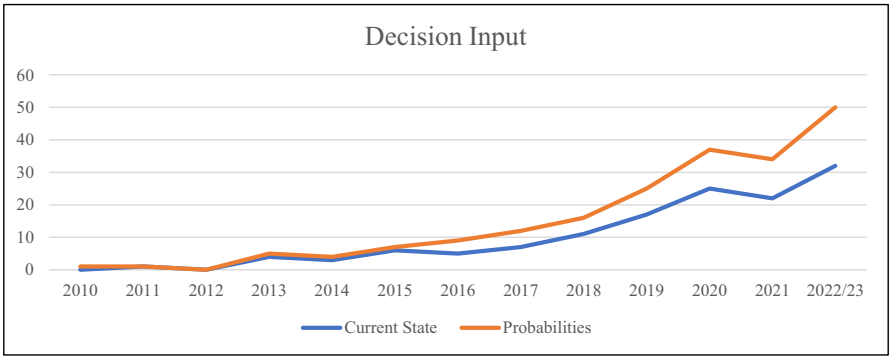
Concepts		References from the literature sample discussing the properties of data
Data type	Structured	[1], [3], [4], [6], [8], [9], [14], [15], [17], [21], [22], [28], [30], [32], [34], [35], [36], [37], [38], [40], [42], [45], [47], [49], [51], [52], [53], [59], [63], [64], [65], [66], [67], [68], [69], [74], [77], [78], [80], [82], [85], [87], [88], [90], [92], [93], [94], [99], [100], [101], [104], [109], [110], [112], [114], [115], [120], [123], [125], [126], [128], [129], [130], [132], [133], [136], [137], [139], [140], [142], [143], [147], [151], [152], [153], [154], [156], [158], [162], [163], [164], [166], [167], [168], [171], [172], [173], [175], [178], [182], [184], [185], [187], [188], [189], [190], [191], [194], [195], [196], [201], [202], [204], [206], [208], [209], [210], [211], [212], [213], [216], [218], [220], [221], [223], [224], [225], [227], [228], [229], [230], [232], [234], [235], [236], [241], [246], [248], [251], [252], [253], [254], [257], [258], [259], [260], [261]
	Unstructured	[1], [3], [12], [21], [30], [34], [42], [45], [47], [48], [52], [54], [64], [66], [67], [69], [70], [78], [80], [85], [86], [87], [90], [92], [93], [99], [104], [109], [113], [117], [118], [125], [129], [130], [132], [137], [143], [147], [151], [153], [163], [167], [179], [182], [187], [188], [189], [190], [195], [202], [204], [209], [210], [211], [212], [214], [220], [223], [224], [227], [234], [235], [249], [258], [261]

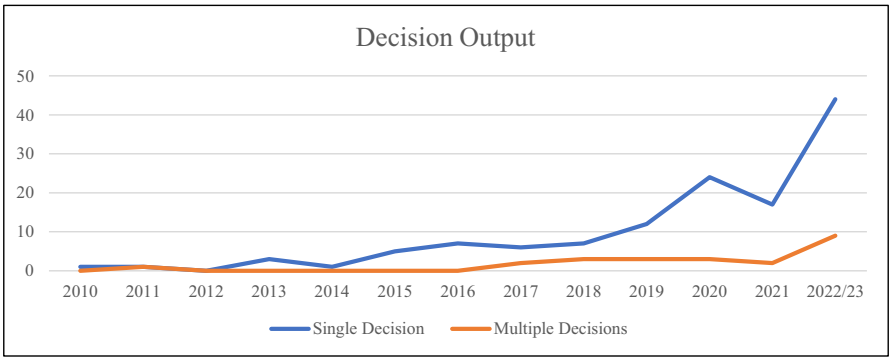
Concepts		References from the literature sample discussing the properties of data
Data velocity	Real-time/streaming	[1], [4], [12], [16], [20], [21], [22], [23], [24], [25], [26], [30], [31], [35], [40], [45], [47], [49], [53], [54], [55], [56], [60], [65], [67], [69], [74], [90], [92], [100], [103], [104], [112], [115], [123], [128], [129], [131], [137], [138], [142], [143], [155], [159], [162], [168], [171], [172], [183], [186], [189], [190], [192], [194], [195], [196], [199], [204], [207], [208], [210], [214], [215], [220], [223], [224], [225], [227], [231], [232], [233], [237], [239], [241], [246], [253], [261]
	Historical/batches	[1], [4], [6], [12], [14], [15], [17], [20], [21], [22], [25], [27], [28], [30], [31], [32], [34], [35], [36], [37], [40], [45], [47], [49], [52], [53], [56], [60], [63], [65], [67], [68], [69], [70], [74], [78], [79], [88], [90], [92], [93], [94], [96], [97], [100], [103], [104], [109], [110], [112], [114], [115], [120], [123], [125], [127], [128], [129], [130], [132], [133], [135], [137], [138], [139], [142], [143], [146], [147], [148], [149], [152], [153], [155], [158], [162], [163], [164], [166], [167], [168], [169], [171], [172], [175], [185], [186], [187], [189], [190], [191], [192], [193], [194], [195], [196], [199], [200], [201], [204], [207], [208], [209], [210], [211], [212], [213], [214], [215], [216], [217], [219], [220], [221], [223], [224], [227], [228], [229], [231], [233], [235], [236], [238], [241], [246], [248], [251], [252], [253], [254], [258], [259], [261]
Data origin	External	[13], [14], [15], [17], [34], [45], [49], [52], [56], [67], [70], [78], [88], [90], [92], [93], [95], [96], [104], [112], [115], [117], [125], [126], [133], [137], [138], [139], [143], [145], [168], [196], [203], [209], [210], [220], [223], [228], [233], [241], [249], [252], [254], [257], [258], [259], [261]
	Internal	[4], [8], [12], [14], [15], [16], [17], [34], [35], [36], [45], [47], [49], [52], [56], [67], [68], [74], [77], [88], [90], [93], [96], [104], [109], [110], [112], [114], [115], [123], [125], [126], [132], [133], [137], [139], [143], [145], [146], [152], [168], [175], [185], [187], [191], [194], [195], [196], [203], [209], [210], [211], [216], [220], [223], [228], [233], [253], [258], [259], [261]

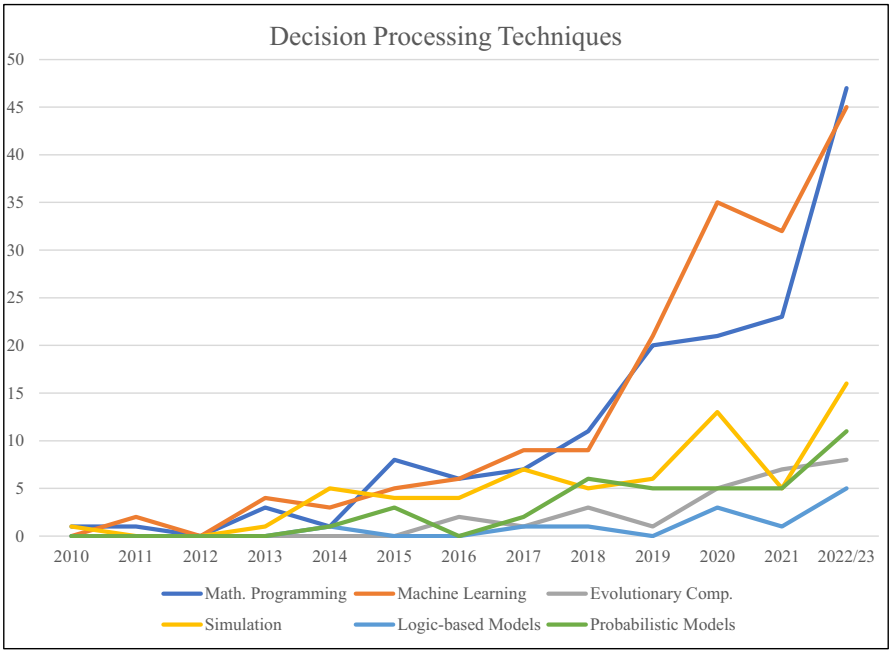
Concepts		References from the literature sample discussing the properties of data
Data generation	Empirical	[1], [3], [4], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [24], [27], [29], [30], [32], [34], [36], [37], [39], [40], [45], [47], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [63], [65], [66], [67], [68], [71], [72], [74], [75], [76], [77], [78], [80], [81], [82], [85], [86], [88], [90], [92], [93], [94], [95], [96], [97], [98], [100], [101], [102], [103], [104], [105], [106], [108], [109], [110], [112], [114], [115], [117], [120], [121], [123], [125], [126], [128], [129], [130], [131], [132], [133], [135], [136], [137], [138], [139], [140], [142], [143], [144], [146], [147], [148], [149], [150], [151], [152], [153], [154], [156], [158], [160], [161], [163], [164], [165], [166], [167], [168], [169], [171], [172], [173], [175], [177], [178], [181], [182], [184], [185], [186], [187], [190], [191], [192], [193], [194], [195], [196], [199], [200], [201], [202], [203], [205], [206], [207], [208], [209], [210], [211], [212], [213], [214], [216], [217], [218], [219], [220], [221], [223], [224], [225], [226], [227], [228], [230], [231], [232], [233], [234], [237], [239], [240], [241], [243], [246], [248], [249], [251], [252], [253], [254], [255], [256], [257], [258], [259], [260], [261], [262]
	Synthetic	[28], [30], [41], [43], [48], [53], [60], [62], [74], [75], [92], [108], [116], [117], [120], [122], [127], [134], [136], [154], [177], [181], [182], [188], [224], [230], [252]
	Assumptions	[1], [11], [20], [27], [32], [52], [56], [65], [66], [77], [80], [83], [90], [92], [100], [104], [113], [116], [118], [123], [125], [127], [130], [134], [137], [144], [156], [166], [190], [200], [202], [218], [224], [228], [231], [239], [246], [255], [258], [260]

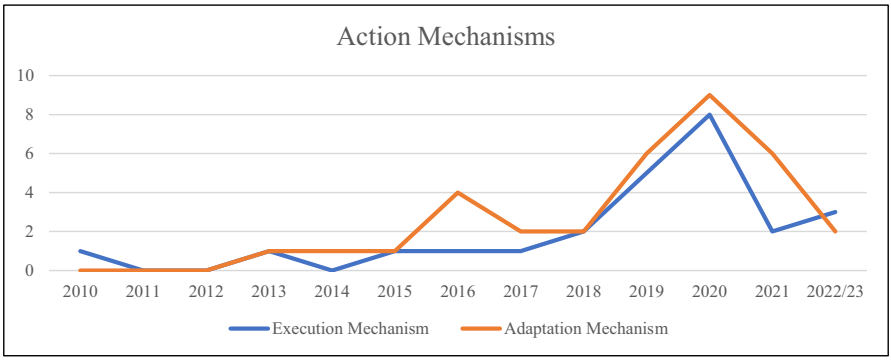
Appendix H: Temporal trend analysis of results

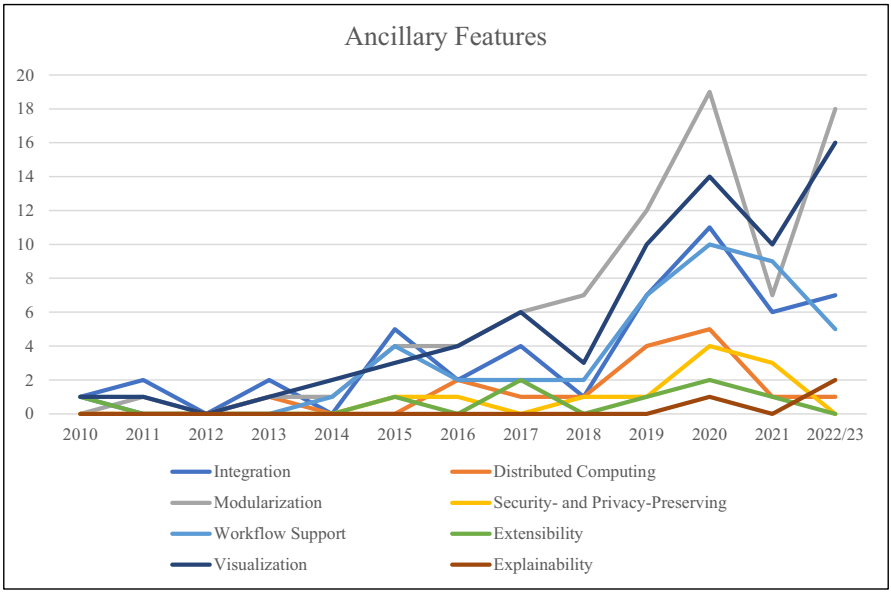


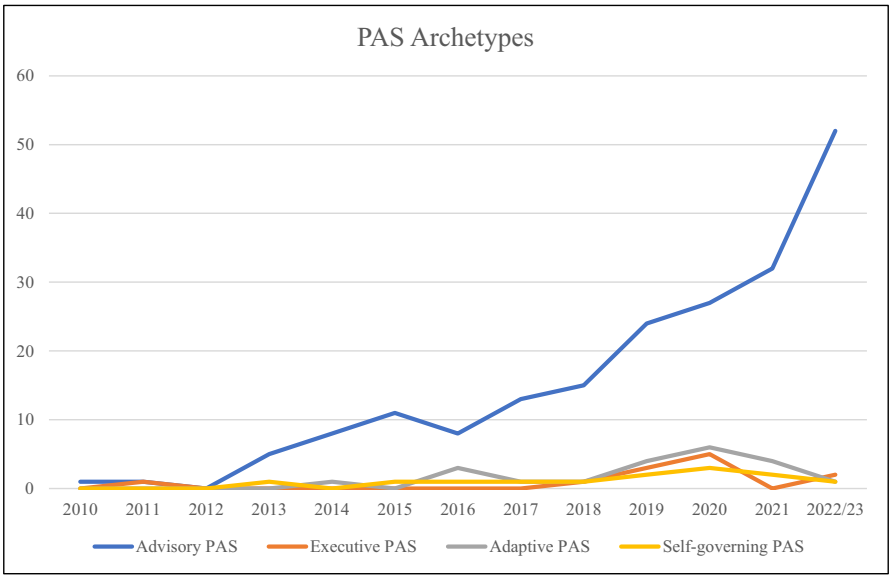

Decision Input
60
50
40
30
20
10
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022/23
Current State
Probabilities

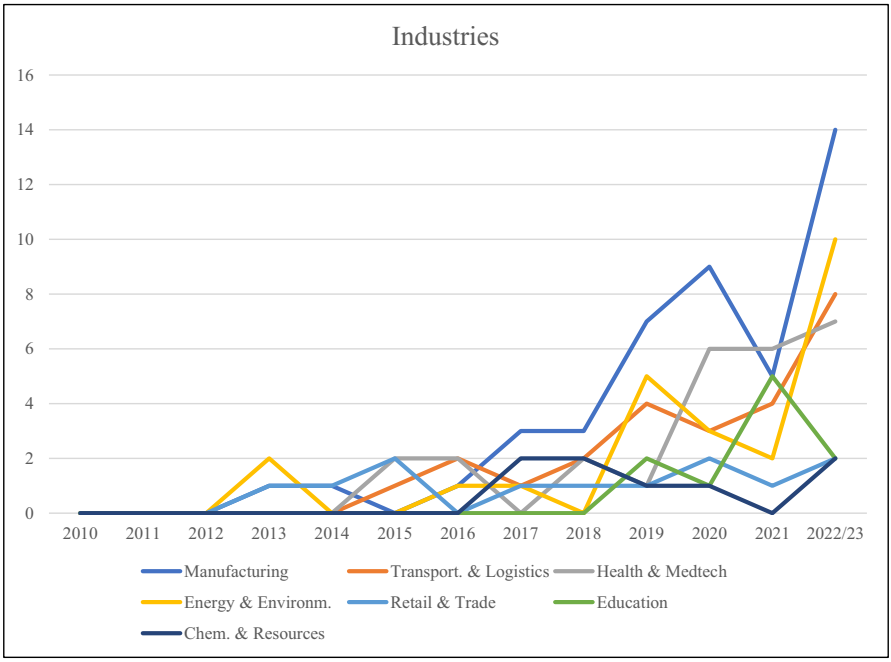

Decision Output
50
40
30
20
10
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022/23
Single Decision
Multiple Decisions


Decision Processing Techniques
50
45
40
35
30
25
20
15
10
5
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022/23
Math. Programming
Machine Learning
Evolutionary Comp.
Simulation
Logic-based Models
Probabilistic Models


Action Mechanisms
10
8
6
4
2
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022/23
Execution Mechanism
Adaptation Mechanism


Ancillary Features
20
18
16
14
12
10
8
6
4
2
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022/23
Integration
Distributed Computing
Modularization
Security- and Privacy-Preserving
Workflow Support
Extensibility
Visualization
Explainability


PAS Archetypes
60
50
40
30
20
10
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022/23
Advisory PAS
Executive PAS
Adaptive PAS
Self-governing PAS



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